Pholders SRS



Software Requirements Specification (SRS) Document
Pholders Rx Creator and Medical Tracker

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Version 1.00

Company Information

1. Document Revisions

Date	Version Number	Document Changes
14/01/2026	0.1	Initial Draft

2. Approvals

Role	Name	Title	Signature	Date
Project Sponsor				
Business Owner				
Project Manager				
System Architect				
Development Lead				
User Experience Lead				
Quality Lead				

Content Lead				

3. Introduction

3.1 Project Summary

3.1.1 Objectives

[These should describe the overall goal in developing the product, high level descriptions of what the product will do, how they are aligned to business objectives, and the requirements for interaction with other systems.]

1. Enhance Patient Safety, Accuracy, and Quality of Care

- Eliminate illegible handwritten prescriptions and reduce medication errors through digital transcription into standardized, verifiable templates with doctor e-signature.
- Prevent adverse drug events via built-in checks (e.g., interactions, allergies, contraindications) when integrated with patient history.
- Improve prescribing quality and formulary adherence (especially relevant in systems like South Africa's medical aids).

2. Improve Medication Adherence and Chronic Disease Management

- Enable precise tracking of medicine intake with personalized schedules, automated reminders, and logging.
- Support better treatment outcomes for chronic conditions (very common in many populations) by making adherence easier and more accountable.
- Allow patients to upload evidence (e.g., photos of generics/alternatives) for personal records and future consultations.

3. Increase Accessibility and Convenience in Healthcare

- Help patients quickly locate nearby doctors, book appointments, and set/get reminders — reducing barriers to care.
- Provide seamless digital prescription delivery to a network of pharmacies, allowing one pharmacy to claim and prepare orders for collection or delivery.
- Enable online medication purchasing with delivery/collection options, minimizing pharmacy wait times and travel.

4. Create a Portable, Unified Patient Medical Record (PHR)

- Serve as a lifelong, patient-owned digital health file storing personal details (e.g., medical aid info), medical history, prescriptions, and records.
- Allow easy searching, printing, emailing, and secure sharing of prescriptions and records with any healthcare provider — eliminating repetitive history-taking.
- Empower patients to own and control their health data across providers, improving continuity of care.

5. Boost Efficiency and Reduce Administrative Burden for Doctors and Pharmacies

- Streamline prescription creation, signing, and transmission for doctors.
- Enable doctors to advertise their practices and reach more patients.
- Generate insightful reports for doctors (e.g., patient adherence trends, practice analytics) and patients (e.g., medication history summaries).

6. Foster Patient Engagement, Trust, and Feedback

- Allow patients to review and rate doctors, building transparency and quality improvement.
- Encourage active patient participation through reminders, tracking tools, and easy access to their own health information.

7. Overall System-Level Goals (Broader Impact)

- Reduce healthcare system inefficiencies (e.g., phone calls, callbacks, lost prescriptions, delays).
- Support cost savings through better adherence (fewer complications/hospitalizations), generic/alternative use, and streamlined processes.
- Promote digital inclusion in healthcare, especially in regions with access challenges (relevant to many African contexts including South Africa).

In essence, the app's overarching objectives are to:

- Make healthcare more patient-empowered, connected, and continuous through a true portable medical record.
- Close the loop from consultation → prescription → dispensing → adherence → follow-up.
- Deliver safer, faster, more convenient, and higher-quality care for patients while creating efficiency gains for providers and pharmacies.

This combination positions the app as both a practical daily tool and a foundational piece of modern, digital-first healthcare — especially powerful in markets moving toward e-prescribing and integrated telehealth/pharmacy ecosystems.

3.1.2 Background

[Provide a brief history of how the project came to be proposed and initiated, including the business issues/problems identified, and expected benefit of implementing the project/developing the product.]

Tobun Professionals has discovered a niche in the market in medicine, where the doctor-to-patient-to-pharmacist business process is causing unreadable scripts, long queues, and multiple round trips. By developing an online prescription maker, scripts will be readable, queues shorter and only one round trip to the pharmacy is guaranteed. The patient will still retain the option to use the traditional process and will have access to the app.

In the traditional system, the doctor writes a handwritten script, gives it to the patient and the patient takes it to the pharmacy to collect medication. Sometimes the medication is not immediately available and so the pharmacy has to order it and the patient has to return later, when the pharmacy sends a notification of the

availability of the medication. Sometimes the pharmacy does not have the exact medication and has to issue a generic while the patient may prefer the prescribed drug.

This system has nurtured a few problems like unreadable scripts, long queues and multiple roundtrips to the pharmacy. Pholders serves to eliminate these issues and track the patient's medical history, prescribed medication and doses.

Pholders aims to function as a comprehensive, patient-centered digital health platform that combines e-prescribing, medication management, telemedicine facilitation, pharmacy integration, and a portable personal health record (PHR) system. Its core vision is to create a lifelong, portable patient medical file — allowing individuals to carry their complete medical history everywhere without starting from scratch at every new healthcare professional, while streamlining the entire journey from consultation to medication adherence and follow-up.

3.1.2.1 Business Drivers

The business drivers for a comprehensive prescription creator, medical tracker, and portable patient health record app (as described) are strong — particularly in a market like South Africa in 2026, where digital health adoption is accelerating, e-prescribing is legally supported (with advanced electronic signatures), chronic disease burdens are high, and private healthcare ecosystems (medical aids, pharmacies, independent doctors) create clear monetization paths.

Here are the key business drivers, grouped logically:

1. Massive Market Pain Points and Unmet Needs

- Poor medication adherence — A global and local issue costing healthcare systems billions (estimates range from $100–300 billion annually worldwide). In South Africa, chronic conditions like diabetes, hypertension, and HIV drive high non-adherence. Features like intake tracking, smart schedules, reminders, and photo uploads for generics create real value and retention.
- Fragmented patient records — Patients repeat history at every visit, wasting time and risking errors. A true portable, patient-owned medical file (PHR) becomes a powerful differentiator, especially with medical aid details stored.
- Illegible/fraudulent prescriptions → Digital transcription + doctor e-signature reduces errors, improves safety, and aligns with SAHPRA regulations and existing platforms (e.g., ICANSCRIPT, EMGuidance Script).
- Access barriers — Long queues, rural/urban divides, and difficulty finding/booking doctors. Doctor search + booking + appointment reminders address this directly.

2. Strong Revenue Model Opportunities

- Transaction fees — Charge doctors/pharmacies per booking, per e-prescription sent/claimed, or per online medication order (common in platforms like RecoMed).
- Subscription/premium tiers — Doctors pay for advertising, advanced reports (adherence analytics, practice insights), priority visibility, or premium features.

- Pharmacy network commissions — Percentage on fulfilled online orders, delivery/collection, or preferred network status.
- B2B partnerships — Medical aids (Discovery, Momentum, etc.) pay to integrate for better chronic care management, formulary checks, cost savings, and member engagement.
- Data & insights (anonymized) — Sell aggregated adherence/trends reports to pharma, medical aids, or researchers.
- Freemium for patients — Core features free to drive massive user growth; monetize via ads, premium reminders, or telehealth add-ons.

3. Macro Trends and Market Growth

- Rising chronic disease & aging population — Driving demand for adherence tools (global medication management apps market growing ~11–12% CAGR to billions by 2030s).
- Digital health boom — Post-COVID acceleration of telehealth, e-prescribing, and mHealth in Africa/South Africa (platforms like RecoMed, HealthDart already scaling with millions in bookings).
- Regulatory tailwinds — E-prescribing legally viable with advanced electronic signatures (Medicines Act + ECTA). SAHPRA supports digital health tools, and private sector adoption is growing fast.
- High smartphone penetration — Makes a mobile-first app highly accessible, even in underserved areas.

4. Network Effects & Competitive Moats

- Two-sided (or multi-sided) platform — More doctors attract more patients → more bookings → more pharmacy integrations → better prices/delivery → more patients → stronger network.
- Data flywheel — Rich patient history + adherence tracking + ratings create valuable insights for all parties, increasing stickiness.
- Patient empowerment & loyalty — Owning a lifelong portable record builds high lifetime value, reducing churn and enabling cross-selling (e.g., wellness, insurance products).

5. Cost Reduction & Value Creation for Stakeholders

- For medical aids/insurers — Better adherence = fewer complications/hospitalizations = lower claims costs.
- For doctors — Less admin, advertising reach, reports for better practice management.
- For pharmacies — Steady digital orders, reduced stockouts via pre-orders, customer loyalty.
- For patients — Convenience, safety, affordability (generics comparison, delivery), peace of mind.

Realistic South African Context (2026 Perspective)In South Africa, similar platforms already exist (e.g., doctor booking like RecoMed, e-scripts like ICANSCRIPT/EMGuidance), but few offer the full closed loop: consultation → digital prescription → pharmacy network → adherence tracking → portable lifelong record.This creates a clear opportunity for differentiation — especially if the app integrates deeply with medical aids, supports Schedule 5–6 medicines legally, and builds trust via strong security/compliance.

Bottom line:

The strongest drivers are:

- Solving adherence + access + continuity in a high-chronic-disease market
- Creating a valuable multi-sided platform with recurring transaction/subscription revenue
- Riding regulatory + digital adoption waves in a private-heavy healthcare system

If executed well (compliance-first, user-friendly, strong partnerships), this type of app has significant potential to become a "must-have" in South African digital health — much like how RecoMed scaled bookings or how GoodRx disrupted affordability globally. If you're building or evaluating this, focus early on regulatory alignment, pharmacy/medical aid integrations, and proven adherence impact to unlock the biggest drivers

3.2 Project Scope

[Describe what work is in scope for the project, and specifically what work is out of scope... beyond the current budget, resources and timeline as approved by the project stakeholders. This is designed to prevent “scope creep” of additional features and functions not originally anticipated.]

3.2.1 In Scope Functionality

The in-scope functionality for the described prescription creator + medical tracker + portable patient medical record app (especially in the South African context as of 2026) can be grouped into core modules that form a closed-loop, patient-centered digital health ecosystem.

This scope focuses on legally compliant e-prescribing (using Advanced Electronic Signatures – AES per Medicines Act + ECTA), medication management, doctor-patient-pharmacy connectivity, and a true patient-owned Personal Health Record (PHR) that travels with the user across providers.

1. Core e-Prescribing & Prescription Management

- Doctor creates/transcribes prescription into a standardized digital template (with required fields: prescriber details, patient ID validation, medicine name/dosage/quantity/instructions, schedule classification).
- Real-time intelligent features (e.g., dosage autofill, basic drug interaction/contraindication alerts based on patient profile).
- Doctor applies legally binding AES (advanced electronic signature, often with OTP/Home Affairs integration) for all schedules of medicines.
- Prescription becomes instantly trackable/auditable in the cloud.
- Patient receives/view/accesses the digital prescription in-app (QR code, secure link, or direct view).
- Patient can search, view history, print, email/share prescriptions securely (with audit trail).

2. Doctor Discovery, Booking & Practice Management

- Geolocation-based search for nearest doctors (filtered by specialty, availability, ratings, medical aid acceptance).
- Online appointment booking with calendar integration.
- Patient sets/receives push notifications & reminders for upcoming appointments.

- Doctors can advertise their practice (profile visibility, promoted listings, basic practice info).

3. Medication Tracking & Adherence (Patient-Facing)

- Automatic import of prescription schedule into personal medication plan.
- Customizable daily/weekly intake schedules with push reminders/notifications.
- Patient logs intake (taken/skipped/missed) with timestamp.
- Upload photos of generics, alternatives, pill bottles, or packaging for personal records/evidence (stored in PHR).
- Basic adherence tracking/reporting (e.g., % adherence over time).

4. Pharmacy Network & Fulfillment Integration

- Patient sends digital prescription to a connected network of pharmacies (real-time transmission).
- Single pharmacy "claims" the prescription (prevents duplicate dispensing).
- Pharmacy prepares order for collection (in-store notification) or delivery (where supported).
- Patient purchases medication online (delivery or collection options) via the app.
- Integration with pharmacy systems for stock visibility (optional/preferred).

5. Portable Personal Health Record (PHR) – The Central Backbone

- Patient-owned, lifelong digital file storing:
  - Personal details (ID, contact, medical aid info/membership).
  - Full prescription history.
  - Medication list (current + past).
  - Uploaded photos/documents (generics, reports, etc.).
  - Basic medical history (self-reported conditions, allergies, chronic illnesses).
- Secure sharing/export (PDF/print/email) to any healthcare professional (no need to start afresh).
- Patient controls access/permissions.
- Searchable archive of all records.

Proposed Category Structure for the PHR

The system should allow dynamic, user-defined categories while providing sensible defaults. Here's a clean, expandable structure:

a. Personal Details (Core Identity & Contact)
   - Full name, ID/passport number, date of birth, gender
   - Contact information (addresses, phone, email — with history of changes)

     Next of kin / emergency contacts

- Marital status, dependents, and basic biographical notes
- Digital identifiers (e.g., tax number, driver's license)

b. Medical Details (Traditional PHR Core)
   - Allergies, chronic conditions, diagnoses
   - Medications (current + history), vaccinations
   - Test results, imaging reports, hospital records
   - Doctor/specialist contacts and visit history
   - Lifestyle data (blood pressure logs, glucose, fitness trackers — if integrated)
   - Advance directives / living will / medical power of attorney

c. Financial Records (Estate & Asset Management — your key requirement)
   - Investments & Shares:
     - JSE shareholdings (company names, share certificate details, broker/transfer secretary info)
     - Unclaimed dividends tracker (link or notes to JSE Claim It portal)
     - Other investments (unit trusts, retirement annuities, stocks on other exchanges)
   - Policies & Insurance:
     - Life insurance policies (policy numbers, insurer contacts, beneficiaries)
     - Medical aid / health insurance
     - Short-term insurance (home, car, etc.)
     - Funeral cover
   - Banking & Accounts:
     - Bank accounts, credit cards, loans
     - Retirement funds / pensions (provident, preservation funds)
   - Other Assets:
     - Property details, vehicles, valuable items
     - Cryptocurrency wallets or digital assets (with secure access notes)
   - Liabilities: Debts, mortgages, etc.

d. Legal & Estate Documents
   - Will / testament (location or secure upload)
   - Letters of executorship (for deceased estates)
   - Power of attorney (general or special)
   - Trust deeds (if applicable)
   - Death certificate template or notes for family

e. Additional/Custom Categories (User can add these)

- Digital Assets — Email accounts, social media, online subscriptions, passwords (via secure vault)
- Family History — Genealogy notes
- Important Documents — Scanned IDs, marriage certificates, title deeds
- Subscriptions & Memberships
- Wishes & Legacy — Funeral preferences, messages to loved ones, organ donation
- Any other user-created category (e.g., "Business Interests", "Pet Care")

The system should support:

- Easy adding/renaming of categories
- Tagging, searching, and filtering across all sections
- Version history and audit trail (who added/edited what)
- Secure file uploads (PDFs of policies, share statements, medical reports)

Access & Inheritance Features (Critical for Financial Records)
For the financial records section especially:

- Tiered access control: The patient (owner) has full control.
- Trusted contacts: Designate 1–3 family members or friends as "legacy accessors".
  - They get read-only (or limited edit) access only after the patient's passing or upon trigger (e.g., medical power of attorney activation).
- Trigger mechanism:
  - Patient can set a "dead man's switch" (e.g., must confirm alive every X months, or via trusted executor notification).
  - Integration with a will or legal document that authorises release.
  - Upon death, the system notifies designated contacts with instructions (e.g., "Here are policy numbers to claim life insurance" or "JSE share details for unclaimed dividends").
- Security best practices:
  - End-to-end encryption
  - Two-factor authentication
  - Granular permissions (e.g., medical details visible to doctors, financial only to family/executor)
  - Compliance with POPIA (Protection of Personal Information Act) in South Africa

This setup directly solves the JSE unclaimed dividends issue: a family member could log in, see exact share details, policy numbers, and contact info, then quickly submit claims via the official JSE Claim It portal (www.jse.co.za/claimit) (www.jse.co.za/claimit) or insurer.

6. Feedback, Ratings & Reporting

- Patients review/rate doctors (post-consultation, star ratings + comments).
- Generate reports:
  - For patients → adherence summary, medication history, upcoming refills.
  - For doctors → practice-level insights (e.g., patient volume, adherence trends – anonymized/aggregated).

Typical In-Scope Summary Table (2026 SA Context)

Module	Key In-Scope Features	Legal/Technical Notes
e-Prescribing	Template creation, AES signing, transmission	Must use AES for validity (SAHPRA compliant)
Doctor Booking	Search, book, reminders	Standard in RecoMed-like platforms
Medication Adherence	Schedules, reminders, intake logging, photo uploads	Core value for chronic care
Pharmacy Loop	Send script → claim → collect/deliver + online purchase	Network-dependent, real-time integration
Portable PHR	Lifelong record, export/share, personal data storage	Patient-controlled, not provider EMR
Ratings & Reports	Doctor reviews, basic analytics	Feedback loop + insights

What is Typically Out-of-Scope (or High-Risk/Phase 2+)

- Full telemedicine/video consultations (separate HPCSA rules & liability).
- AI-based diagnosis or automated prescribing.
- Direct integration with public sector systems (e.g., Ideal Clinic/Hospital EMRs).
- Advanced clinical decision support (e.g., full formulary + medical aid real-time claim adjudication).
- Handling of Schedule 6+ controlled substances without extra verification layers.
- Wearable/device auto-sync (e.g., blood pressure from smartwatch) – possible but adds complexity/compliance.
- Full EHR-level features (provider charting, billing codes).

This scope delivers the complete vision — consultation → legal digital script → pharmacy fulfillment → adherence tracking → portable lifelong record — while staying within realistic regulatory boundaries (AES mandatory for e-scripts, patient consent for data sharing, POPIA compliance).If built this way, the app becomes a powerful, differentiated player in South Africa's private digital health space alongside tools like EMGuidance Script, ICANSCRIPT, and RecoMed — but with the unique full-loop + true PHR advantage.

3.2.1.1 Project Phases

PHASE 1

- Search for nearest doctor and book appointment. Once the patient is on the system, and visits doctor, doc should be able to create the script and patient should be able to pay for the consultation prior to entering doctor's office using self payment online. When payment is confirmed or medical aid payment is applicable app will release script to patient and pharmacies.
- Allow patient to find doctor nearest to them (if under medical aid, doctor that’s covered by medical plan) using geolocation.
- Allow patient to review doctor’s service.
- Allow patient to set/receive reminders on appointments.
- Establish regulatory partnership e.g. AES signing (Refer to section 3.3.1 Dependencies)
- Establish Pharmacy Partnership Network → Agreements with chains/independents to claim prescriptions + handle orders/delivery. Also develop pharmacy integrations.
- Medical Aid Switches

PHASE 2

- Include shopping cart for over the counter medication for collection only.
- Incorporating courier services for deliveries - even for prescription medication. The system should also get updates from pharmacy of prescription medication and generics that need to be refilled.
- Ability to upload pic of generics or alternative medicine for own records and evidence.
- Engage doctors to advertise (to make patients more likely to book) to get new patients and keep

existing patients

  - doctor can subscribe to push advertisements of treatments they do and specials etc. stuff that needs to be done on a regular basis - wellness checks, BPs etc
  - when a patient searches for a nearby doctor they will receive push notifications of the doctor’s ads

PHASE 3

- Medical records (Portable Personal Health Record) must be accessible to any doctor according to the patient’s wishes. When the patient makes a booking he is agreeing to share his medical records and

personal details with that doctor.

- Doctor can request patient's medical records and patient can give permission.
- Medication tracking and uploading of pictures of medications and schedule when to use

them.

3.2.2 Out of Scope Functionality

The out-of-scope functionality for the described prescription creator + medical tracker + portable patient medical record app (in the South African context as of January 2026) refers to features that would typically not be included in an MVP or standard version of such a patient-facing platform. This is due to:

- High regulatory/compliance risk (HPCSA Telehealth Guidelines, SAHPRA rules, Medicines Act requirements for e-prescribing with AES, POPIA data protection).
- Increased clinical liability for the app provider.
- Scope creep beyond a patient-owned Personal Health Record (PHR) into full provider-controlled Electronic Health Record (EHR) territory.
- Technical, legal, or operational complexity that requires separate licensing, integrations, or approvals.

High-Risk / Regulatory Out-of-Scope Features

- Full telemedicine / video or real-time consultations — The app focuses on post-consultation tools (e-prescribing, booking, tracking). Direct virtual doctor-patient diagnosis, treatment initiation, or prescribing based solely on questionnaires/online forms is explicitly cautioned against by HPCSA (not an acceptable standard of care without prior relationship in many cases).
- AI-driven diagnosis, automated prescribing, or clinical decision support — Features like symptom checkers, AI suggesting medicines/dosages, or automated alerts for serious conditions (beyond basic interaction checks) risk classifying the app as a medical device requiring SAHPRA registration.
- Prescribing or handling of high-schedule controlled substances (e.g., Schedule 6+ narcotics) without extra layers — While e-prescribing with AES is allowed, apps often avoid direct handling of highly controlled meds due to strict verification/pharmacist burden.
- Direct real-time claim adjudication or billing integration with medical aids — Full automated pre-authorization, real-time benefit checks, or claims submission often requires deep B2B partnerships and provider-side systems.

Provider-Side / Clinical Workflow Features (EHR Territory)

- Full provider charting, progress notes, or encounter documentation — This belongs in doctor-controlled EHR/EMR systems (e.g., practice management software), not a patient-owned PHR.
- Integration as primary EMR/EHR for clinics/hospitals — The app is patient-controlled and portable; it shouldn't serve as the official legal clinical record for providers.
- Advanced clinical monitoring (e.g., real-time vital signs from wearables, remote patient monitoring for acute/high-risk conditions like unstable diabetes or heart failure) — High-risk scenarios require physical exams and fall outside safe telehealth boundaries.

Other Common Out-of-Scope Items

- Public sector / state hospital system integrations (e.g., direct pull from Ideal Clinic or provincial EMRs) — Due to interoperability gaps, data sovereignty, and security concerns.
- Nurse-initiated or expanded prescriber support — The app is doctor-focused for prescribing; nurse/pharmacist prescribing requires separate SAHPRA/HPCSA authorization.
- Cross-border / international prescribing — Complicated by HPCSA registration requirements for foreign practitioners.
- Advanced analytics or monetization of identifiable patient data — Strict POPIA limits; anonymized aggregated insights might be in-scope, but anything identifiable is high-risk.
- Physical device integrations (e.g., smart pill dispensers, auto-sync with blood pressure monitors) — Adds hardware compliance and overestimates adherence (e.g., opening ≠ ingestion).
- Emergency / urgent care features (e.g., auto-escalation to ambulance) — Requires 24/7 clinical oversight and liability coverage.

Quick Summary Table (South Africa 2026 Context)

Category	Typically Out-of-Scope Examples	Main Reason(s)
Telemedicine / Virtual Care	Video consultations, AI diagnosis, first-time prescribing	HPCSA guidelines, liability, acceptable standard of care
Advanced Clinical Tools	Full decision support, wearable auto-monitoring, acute alerts	SAHPRA medical device risk, high clinical liability
Provider EHR Features	Charting, progress notes, hospital EMR integration	PHR vs EHR distinction; provider control required
High-Controlled / Specialized	Schedule 6+ direct handling, nurse prescribing support	Strict verification & authorization rules
Billing / Claims	Real-time medical aid adjudication, automated claims	Complex B2B integrations, regulatory hurdles
Emergency / High-Risk	Real-time vital monitoring, urgent escalation	Requires physical exam & 24/7 oversight

Bottom line: The app stays powerful and differentiated by focusing on the closed loop of legal e-prescribing (doctor-signed AES) → pharmacy network → patient adherence tracking → portable PHR, without crossing into high-liability clinical delivery or full EHR territory. Features like these would require separate regulatory approval, much higher liability insurance, and often a shift from patient-facing to provider/clinic software.

This keeps the app compliant, scalable, and focused on its core value: empowering patients with a lifelong, portable medical file while streamlining private-sector workflows. If you're planning phases, many of these could become Phase 2+ add-ons with partnerships (e.g., HPCSA-approved telehealth module).

3.3 System Perspective

[Provide a complete description of the factors that could prevent successful implementation or accelerate the projects, particularly factors related to legal and regulatory compliance, existing technical or operational limitations in the environment, and budget/resource constraints.]

3.3.1 Dependencies

The dependencies of a prescription creator + medical tracker + portable patient medical record app (as described, in the South African context as of January 2026) span legal/regulatory, technical, integrations/partnerships, security/compliance, and operational categories.

These are essential for the app to function legally, securely, scalably, and with real-world utility — especially given the requirement for legally valid e-prescribing (via Advanced Electronic Signatures – AES), a closed-loop pharmacy network, and a true patient-owned Personal Health Record (PHR).

1. Regulatory & Legal Dependencies (Critical – Non-Negotiable)

- Medicines and Related Substances Act (Act 101 of 1965) + Regulations (esp. Regulation 33) → Requires prescriptions to be generated by an "electronic agent" and signed with AES for legal validity across all schedules (including S5/S6).
- Electronic Communications and Transactions Act (ECTA, 2002) → Defines AES requirements (accredited by South African Accreditation Authority – SAAA; usually involves face-to-face ID verification + digital certificate + OTP/2FA).
- HPCSA Ethical Guidelines (esp. Booklet 10 on Telehealth, updated ~2021–2022) → Governs remote interactions, patient consent, record-keeping, and boundaries (e.g., no full diagnosis solely via app without established relationship in many cases).
- POPIA (Protection of Personal Information Act) → Strict rules for handling sensitive health data (consent, minimization, security, breach notification).
- SAHPRA oversight → For any features bordering on medical device classification (e.g., advanced drug interaction checks).

Without AES integration from an accredited provider (e.g., TrustFactory, DocuSign accredited services), digital prescriptions are not legally valid in SA.

2. External Service & Integration Dependencies

- AES Certificate Provider → E.g., TrustFactory or similar accredited authentication service (doctors must obtain personal AES certificates; app must integrate signing flow with OTP/2FA verification).
- Pharmacy Network & Dispensing Systems → Real-time prescription claiming/transmission to connected pharmacies (e.g., via APIs from systems like Perfect Pharmacy Manager, ProPharm, Rx30, or emerging platforms like EMGuidance Script / PrescriBIT).

- Medical Aid Switches → For real-time benefit checks, co-payment info, formulary validation (major switches: Interpharm/Medscheme, Mediswitch, Medikredit, Zieto — covering ~150+ schemes like Discovery, Momentum).
- Geolocation & Mapping Services → Google Maps API, Mapbox, or OpenStreetMap for "nearest doctor" search + directions.
- Push Notification & Messaging → Firebase Cloud Messaging (FCM), OneSignal, or local SMS gateways (e.g., Clickatell, Twilio) for reminders (appointments, intake).
- Payment Gateway (for online medication purchases) → PayFast, Peach Payments, or Yoco (PCI-DSS compliant for card payments + delivery integration).
- Email/SMS Delivery → For prescription sharing/printing (e.g., SendGrid, Amazon SES).

3. Technical & Infrastructure Dependencies

Common/popular stack for SA health apps in 2025–2026 (based on scalability, cost, and compliance needs):

Layer	Recommended Dependencies / Technologies	Purpose / Notes
Frontend (Mobile)	React Native / Flutter	Cross-platform iOS + Android (high adoption in SA health apps)
Frontend (Web/Admin)	React.js / Vue.js + TypeScript	Doctor portal, patient dashboard
Backend	Node.js (Express) / Python (Django/FastAPI)	API-driven, scalable; Python strong for data/health logic
Database	PostgreSQL (with encryption) / MongoDB	Structured health data + documents/photos
Cloud Hosting	AWS (Cape Town region), Azure, Google Cloud	POPIA-aligned data residency in SA/Africa
File Storage	AWS S3 / Google Cloud Storage + encryption	Secure photo uploads (generics, evidence)
Authentication	Firebase Auth / Auth0 + AES integration	Secure login + doctor AES signing
Real-time	WebSockets (Socket.io) / Firebase	Reminders, prescription status updates

- HL7 / FHIR (optional but increasingly expected) → For future interoperability with hospital EMRs or national systems.
- Barcode/QR Code Generation → Libraries like ZXing or qr-code-styling for prescription QR codes.

4. Security & Compliance Dependencies

- Encryption at rest/transit (AES-256).
- Audit logging for every prescription/action (immutability for legal defense).
- Regular penetration testing + vulnerability scanning.
- Backup & disaster recovery (POPIA requirement).
- Consent management engine (granular patient permissions for sharing PHR).

5. Operational & Business Dependencies

- Doctor Onboarding Process → Verification (HPCSA registration check) + AES certificate acquisition support.
- Pharmacy Partnership Network → Agreements with chains/independents to claim prescriptions + handle orders/delivery.
- Patient Verification → ID validation (e.g., via Home Affairs API or trusted KYC providers) for medical aid linkage + security.

Bottom line for South Africa 2026: The heaviest dependencies are AES signing infrastructure (legal make-or-break), medical aid switch APIs (for real value in private sector), and pharmacy system integrations (for the closed-loop dispensing). Without these, the app cannot deliver legally valid e-prescriptions or seamless fulfillment — the core differentiator.

A realistic MVP would start with AES + basic pharmacy push (e.g., secure link/QR), then expand to full network + medical aid checks. Focus early on accredited AES + POPIA compliance to avoid shutdown risks.

If you're planning development, I recommend prioritizing regulatory partnerships first — many SA health apps fail here.

3.3.2 Assumptions

The assumptions underlying the design, development, and successful operation of a prescription creator + medical tracker + portable patient medical record app (as described) in the South African context (as of January 2026) are critical to recognize. These are implicit beliefs about the environment, users, technology, regulations, and market that — if proven false — could significantly impact viability, adoption, or compliance.

Here are the key assumptions, grouped by category:

1. Regulatory & Legal Assumptions

- E-prescribing with Advanced Electronic Signatures (AES) remains legally valid and accepted across all medicine schedules (including S5/S6) under the Medicines Act + ECTA, with platforms using accredited AES providers (e.g., TrustFactory-integrated flows with OTP/2FA) being recognized as bona fide by SAHPRA, pharmacists, and courts.
- Pharmacists will accept and dispense digital prescriptions transmitted via the app (with AES verification visible via seals, indicators, or QR codes) without requiring additional wet-ink signatures or physical verification in most cases.

- No major regulatory changes (e.g., new SAHPRA restrictions on patient-owned apps handling prescriptions or stricter telehealth boundaries) will block the core e-prescribing + PHR features in the near term.
- POPIA compliance (patient consent, data minimization, breach protocols) is sufficient to avoid enforcement actions when storing/sharing sensitive health data in a patient-controlled manner.

2. Market & Adoption Assumptions

- Private-sector focus dominates viability — Doctors, pharmacies, and patients in the private healthcare ecosystem (medical aids like Discovery/Momentum, independent practices, retail chains) will adopt the app, as public-sector integration remains fragmented and slow (due to infrastructure gaps, interoperability issues with HNSF, and low digital maturity).
- Sufficient critical mass / network effects will emerge — Enough doctors will onboard (with AES certificates), pharmacies will join the network for claiming/dispensing, and patients will use the app consistently to create value (e.g., seamless fulfillment, adherence tracking).
- Patients (especially chronic care users) will actively engage with features like intake logging, reminders, photo uploads, and portable PHR sharing, assuming reasonable smartphone access, digital literacy, and trust in data privacy/security.
- Doctors will see value in advertising, reports, reduced admin, and patient ratings outweighing any perceived liability or workflow disruption.

3. Technical & Infrastructure Assumptions

- Reliable internet/connectivity is available for key users (urban/suburban private patients, doctors, pharmacies) for real-time prescription transmission, reminders, and PHR access (offline fallbacks are possible but limited).
- Integration feasibility — APIs or secure channels exist (or can be negotiated) with medical aid switches (for benefit/formulary checks), pharmacy systems (for claiming), and AES providers without prohibitive cost/delays.
- Smartphone penetration & app usage remains high enough among target users (private-sector patients, employed uninsured "missing middle") to drive scale.

4. User Behavior & Value Assumptions

- Adherence tracking improves outcomes — Patients will log intake accurately, respond to reminders, and use photo evidence, leading to measurable better chronic disease management (justifying retention and partnerships).
- Portable PHR reduces repetition — Healthcare professionals (especially in private sector) will accept/view/share the patient-owned record (PDFs, exports) as reliable, reducing "start afresh" issues.
- Patients value convenience over privacy concerns — Features like online purchase/delivery, pharmacy claiming, and doctor ratings will drive usage despite general data sensitivity in health.

5. Business & Economic Assumptions

- Revenue streams are realistic — Transaction fees (bookings, prescriptions, orders), doctor subscriptions (advertising/reports), pharmacy commissions, or medical aid partnerships will cover costs and enable

profitability.

- Competition & differentiation — Existing players (e.g., EMGuidance Script, ICANSCRIPT, RecoMed) won't fully replicate the closed-loop + true lifelong PHR model quickly, or partnerships can be formed instead of direct competition.
- Cost of compliance & development — Building AES integration, POPIA-grade security, audit trails, and scalable cloud infrastructure (AWS/Azure in SA region) remains affordable within private-sector funding models.

Quick Summary Table of Major Assumptions (South Africa 2026 Reality Check)

Category	Key Assumption	Risk Level if Wrong	Mitigation Notes
Regulatory	AES e-scripts fully accepted & stable	High	Monitor SAHPRA/HPCSA updates closely
Adoption (Private Sector)	Network effects kick in with doctors/pharmacies	High	Start with pilots in high-density areas
Infrastructure	Adequate connectivity for real-time features	Medium	Build robust offline capabilities
User Behavior	Patients actively maintain/use PHR & tracking	Medium-High	Focus on UX, education, incentives
Economic	Monetization covers high compliance/tech costs	Medium	Validate early with B2B partnerships

Bottom line: The app's success heavily assumes a private-sector-dominant, digitally-maturing ecosystem with stable e-prescribing rules (AES-based), willing network participants, and engaged chronic-care patients who trust and actively use a patient-owned portable record.

Many of these assumptions hold reasonably well in urban/private contexts (where tools like EMGuidance Script/ICANSCRIPT already operate), but they become riskier in broader/public or rural rollout. Validating these early (via regulatory opinions, pilot adoption data, user interviews) is essential to avoid building on shaky foundations.

3.3.3 Constraints

The constraints of building and operating a prescription creator + medical tracker + portable patient medical record app (as described) in South Africa (as of January 2026) are significant. They stem primarily from strict

regulatory frameworks, practical adoption barriers, technical/security limitations, and operational realities in the private healthcare sector.

These constraints limit scope, increase costs/complexity, slow rollout, and create ongoing compliance risks — even though e-prescribing with Advanced Electronic Signatures (AES) is legally established and growing (e.g., via platforms like EMGuidance Script).

1. Regulatory & Legal Constraints (Most Binding)

- Mandatory AES for legal validity — All digital prescriptions (across all schedules, including S5/S6) must be generated by an "electronic agent" and signed with a legally accredited AES (ECTA + Regulation 33 of Medicines Act). This requires integration with accredited providers (e.g., TrustFactory), OTP/2FA flows, and visible verification (e.g., seals). Non-AES scripts (e.g., simple digital PDFs without proper signing) remain invalid — pharmacists may refuse them, especially for higher schedules.
- No first-time prescribing via app-only/questionnaire — HPCSA Telehealth Guidelines (updated ~2021–2022, still current in 2026) state that treatment/prescribing based solely on online forms/questionnaires is not an acceptable standard of care. The app cannot enable full remote diagnosis or initial prescribing without an established doctor-patient relationship and proper assessment (often requiring in-person elements).
- Schedule 5 & 6 restrictions — These high-risk/controlled substances face extra scrutiny. While AES e-scripts are allowed, some pharmacists still prefer direct-from-doctor transmission (not patient-mediated) to prevent duplication/alteration. Originals for S6 often require tight controls (e.g., 30-day validity, no repeats in many cases).
- POPIA (strict health data rules) — Health information is "special personal information" — processing is prohibited by default unless specific conditions met (e.g., patient consent, medical necessity). Constraints include:
  - Granular consent management.
  - Strict data minimization, security (encryption, audits), breach reporting.
  - Limits on cross-border transfers (must have equivalent protection).
  - High risk of fines/imprisonment for breaches; patient-owned PHR must allow easy deletion/access rights, but conflicts with retention needs for continuity of care.
- No full telemedicine integration — The app cannot include built-in video consultations or AI diagnosis without separate HPCSA approval — it must remain post-consultation focused.

2. Practical & Adoption Constraints

- Network effects & critical mass required — Limited value until enough doctors (with AES certificates) and pharmacies join for seamless claiming/dispensing. Many pharmacies still resist patient-mediated digital scripts (prefer direct transmission) due to authentication concerns.
- Doctor onboarding barriers — Doctors need HPCSA registration verification + AES certificate (costly/time-consuming process via accredited providers). Low adoption in some segments due to workflow changes, perceived liability, or preference for legacy systems.
- Pharmacist verification burden — Pharmacists must authenticate every e-script (Regulation 33(4)) — app must provide clear AES indicators/seals, but resistance persists for higher schedules or patient-forwarded

scripts.

- Infrastructure & connectivity gaps — Uneven internet in rural/underserved areas limits real-time features (e.g., prescription transmission, reminders). Offline fallbacks are possible but reduce usability for core loop.

3. Technical & Security Constraints

- High compliance costs — AES integration, end-to-end encryption, immutable audit logs, POPIA-grade security (e.g., regular pen-testing), and data residency (preferably SA/Africa cloud) add significant development/maintenance expense.
- Interoperability limitations — No mandatory national standards for full medical aid switch integration (real-time benefits/formulary) or public-sector EMRs — limits seamless claims/adjudication.
- Liability exposure — App provider faces risks if features contribute to errors (e.g., bad adherence tracking leading to harm) — requires strong disclaimers, insurance, and avoidance of clinical decision support.

Quick Summary Table of Major Constraints (South Africa 2026 Perspective)

Category	Key Constraint Example	Impact Level	Workaround/Notes
Regulatory (AES)	Must use accredited AES + OTP for all e-scripts	Very High	Integrate TrustFactory-like providers
Regulatory (HPCSA/POPIA)	No questionnaire-only prescribing; strict health data rules	High	Stay post-consultation; granular consents
Higher Schedules (S5/S6)	Extra verification; potential pharmacist resistance	High	Focus on lower schedules initially
Adoption/Network	Needs doctors/pharmacies to onboard & use consistently	High	Start urban/private pilots
Technical/Security	Encryption, audits, SA data residency required	Medium-High	Use compliant cloud (AWS Cape Town)
Infrastructure	Connectivity gaps in rural areas	Medium	Robust offline modes + SMS fallbacks

Bottom line: The app can legally and powerfully deliver a closed-loop e-prescribing + adherence + portable PHR experience in the private sector — but only if built AES-first, POPIA-compliant, and focused on post-consultation tools (no diagnosis/prescribing initiation). The biggest constraints are AES dependency, pharmacist acceptance for higher schedules, strict POPIA controls on health data, and building network effects in a fragmented ecosystem.

Many existing platforms (e.g., EMGuidance Script) succeed by staying narrowly compliant — your app's full vision is achievable but requires regulatory partnerships, phased rollout (start with lower schedules/urban focus), and strong legal advice to navigate ongoing risks.

3.3.4 Risks

The risks associated with developing and operating a prescription creator + medical tracker + portable patient medical record app (as described) in South Africa (January 2026 context) are substantial and multi-layered. They arise from regulatory non-compliance, cybersecurity vulnerabilities, clinical/patient safety issues, adoption barriers, and financial/operational exposures.

These risks are amplified by the app's handling of sensitive health data (classified as "special personal information" under POPIA), legally binding e-prescribing (requiring AES), and the creation of a patient-owned lifelong PHR that must be portable yet secure.

1. Regulatory & Legal Risks (Highest Priority)

- Non-compliance with e-prescribing rules — If AES integration fails (e.g., improper OTP/verification, missing seals/indicators), prescriptions are invalid → pharmacists refuse dispensing → legal challenges from SAHPRA/HPCSA, fines, or app shutdown. Higher schedules (S5/S6) face extra scrutiny and resistance.
- POPIA violations — Mishandling health data (e.g., inadequate consent, poor encryption, unauthorized sharing, cross-border transfers without safeguards) → complaints to Information Regulator → administrative fines up to R10 million, civil claims for damages, or criminal prosecution (up to 10 years imprisonment). Recent breaches (e.g., NHLS 2024 ransomware) highlight enforcement focus.
- HPCSA & professional liability — Doctors risk sanctions if app features lead to perceived unprofessional conduct (e.g., inadequate verification of patient relationship before prescribing). App provider could face indirect liability under CPA or MRSA if classified as enabling unsafe practices.
- Potential medical device classification — If advanced features (e.g., interaction alerts, adherence analytics) influence decisions, SAHPRA may classify app as SaMD → mandatory registration/licensing → delays/costs.

2. Cybersecurity & Data Breach Risks (Very High in 2026 SA Context)

- Ransomware & data theft — Healthcare is prime target; NHLS 2024 attack disrupted labs, deleted backups, exposed terabytes of data. Similar breach in your app could expose millions of records (prescriptions, medical aid, photos, adherence logs) → identity theft, blackmail, stigma (especially chronic/HIV conditions).
- Third-party/vendor vulnerabilities — Integrations (AES providers, pharmacies, medical aids) introduce risks; legacy/outdated systems common in SA healthcare → supply-chain attacks.
- Phishing & insider threats — Low digital literacy among users/doctors → credential compromise → unauthorized access to PHRs.
- Consequences — Massive reputational damage, loss of trust (patients abandon app), mandatory breach notifications, regulatory investigations, class-action lawsuits.

3. Patient Safety & Clinical Risks

- Medication errors/adherence issues — If app mis-transcribes, fails reminders, or allows photo uploads without validation → wrong intake → adverse events (hospitalization/death) → malpractice claims against doctors (and potentially app).
- Over-reliance on digital tools — Patients skip consultations or self-adjust based on app data → delayed care, especially chronic conditions.

- Fraud/abuse — Fake/altered digital scripts (despite AES) → illegal dispensing → public health risks (e.g., antibiotics resistance from street sales).

4. Adoption & Operational Risks

- Network failure / low critical mass — Insufficient doctors (with AES) or pharmacies → poor user experience → churn → failure to achieve value (e.g., no seamless claiming/delivery).
- Pharmacist resistance — Preference for direct transmission over patient-mediated scripts → delays/refusals → negative reviews.
- Infrastructure limitations — Uneven connectivity (rural areas) → failed real-time features → safety risks.

5. Financial & Reputational Risks

- High compliance costs → AES integration, POPIA-grade security (encryption, audits, pen-testing), insurance → could make app unprofitable.
- Litigation & insurance → Claims from breaches/errors → high premiums or uninsurable.
- Reputational collapse → One major breach (like NHLS) → media storm → user exodus, partner withdrawals.

Quick Summary Table of Key Risks (South Africa 2026 Perspective)

Risk Category	Primary Examples	Severity	Potential Impact
Regulatory/Legal	POPIA breach, invalid AES scripts, HPCSA sanctions	Very High	Fines R10m+, shutdown, criminal charges
Cybersecurity/Data Breach	Ransomware, vendor exploits, NHLS-style incidents	Very High	Mass data exposure, identity theft, lawsuits
Patient Safety/Clinical	Errors in transcription/adherence, fraud	High	Harm to users, malpractice claims
Adoption/Operational	Low network effects, pharmacist resistance	High	Churn, failure to scale
Financial/Reputational	Compliance costs, breach fallout	Medium-High	Insolvency, permanent brand damage

Bottom line: The app's ambitious scope (legal e-prescribing + closed-loop fulfillment + true portable PHR) creates high-reward potential but also extreme risks — especially in a market with rising cyberattacks (healthcare breaches surged), strict POPIA enforcement, and pharmacist skepticism toward patient-forwarded scripts.

Mitigation requires regulatory-first design (AES mandatory, POPIA officer, legal opinions), robust security (zero-trust, regular audits, insurance), phased rollout (start urban/private, lower schedules), and strong

disclaimers/partnerships (doctors/pharmacies/medical aids).Without these, the risks could easily outweigh benefits — many similar initiatives have struggled or failed due to compliance/security issues.

3.3.5 Issues

The issues (challenges, limitations, and practical problems) with developing and operating a prescription creator + medical tracker + portable patient medical record app (as described) in South Africa (as of January 2026) are significant. They stem from regulatory hurdles, pharmacist/practitioner resistance, adoption barriers, technical limitations, and persistent cybersecurity vulnerabilities in the healthcare sector.

These issues make full implementation difficult, slow, and risky — even though platforms like EMGuidance Script, ICANSCRIPT, and PrescriBIT have made progress with AES-based e-prescribing.

1. Regulatory & Legal Issues

- Strict requirements for legal validity — All e-prescriptions must use Advanced Electronic Signatures (AES) with proper verification (OTP/2FA, accredited providers like TrustFactory). Patient-mediated scripts (sent to patient then to pharmacy) often face extra scrutiny or rejection, especially for higher schedules (S5/S6), as pharmacists must authenticate to avoid dispensing without a valid prescription.
- Pharmacist authentication burden — South African Pharmacy Council (SAPC) strongly recommends direct transmission from doctor to pharmacy. Patient-forwarded scripts (e.g., via app) require additional validation, raising concerns about alteration, duplication, or prior dispensing — particularly amid rising prescription medicine addiction concerns.
- HPCSA telehealth boundaries — The app cannot support full remote diagnosis, first-time prescribing, or questionnaire-only consultations without an established doctor-patient relationship. Treatment/prescribing based solely on online data is not acceptable standard of care.
- POPIA complexities — Handling sensitive health data (prescriptions, medical aid details, adherence logs, uploaded photos) requires granular consent, strict minimization, encryption, and breach protocols. Cross-border data transfers (e.g., cloud services) face uncertainty due to unclear enforcement mechanisms.

2. Pharmacist & Practitioner Acceptance Issues

- Resistance to patient-mediated digital scripts — Many pharmacists prefer direct doctor-to-pharmacy transmission for security and traceability. Patient-presented scripts (via app link/QR) create extra workload and risk of refusal, especially for controlled substances.
- Low overall e-prescribing adoption — Despite legal support, implementation remains slow in both private and public sectors due to knowledge gaps, negative perceptions, and outdated systems. Pharmacists show moderate knowledge but significant barriers in software, regulations, and workplace fit.
- Doctor onboarding friction — Requires HPCSA verification + costly/time-consuming AES certificates, plus workflow changes — leading to low participation.

3. Practical & Operational Issues

- Network effects & critical mass problem — The closed-loop (doctor → app → pharmacy claiming → fulfillment) only works with widespread adoption. Limited participating doctors/pharmacies → poor user experience → low retention.

- Infrastructure & connectivity gaps — Uneven internet in rural/underserved areas disrupts real-time features (prescription transmission, reminders, tracking).
- Interoperability limitations — No seamless national standards for medical aid switches (real-time benefits/formulary) or public-sector systems — restricts value in private-heavy ecosystem.
- User behavior challenges — Patients may not consistently log intake, upload evidence, or maintain PHR — reducing adherence benefits.

4. Cybersecurity & Data Protection Issues

- High vulnerability of healthcare apps — South Africa's healthcare sector faces frequent attacks (ransomware, data breaches). Recent incidents (e.g., NHLS 2024 ransomware stealing 1.2TB patient data, disrupting services for months) highlight risks to apps storing sensitive records.
- Legacy & third-party risks — Reliance on outdated systems or vendors introduces vulnerabilities (e.g., exposed medical devices rank SA high globally). Third-party integrations (AES, pharmacies, medical aids) create supply-chain weaknesses.
- Breach consequences — Potential mass exposure of prescriptions, medical aid info, adherence data → identity theft, stigma (e.g., chronic conditions), regulatory fines (up to R10m), lawsuits, and total loss of user trust.

Summary Table of Key Issues (South Africa 2026 Context)

Issue Category	Main Problems & Examples	Severity / Impact Level
Regulatory/Legal	AES mandatory + pharmacist authentication burden; patient-mediated resistance; POPIA gaps	Very High (legal invalidity, fines, shutdown risk)
Acceptance/Adoption	Pharmacist preference for direct scripts; low doctor onboarding; network effects needed	High (limited usability without scale)
Operational/Practical	Connectivity gaps; interoperability limits; user engagement low	Medium-High (disrupted features, churn)
Cybersecurity	Ransomware/data breaches common (NHLS example); vendor risks	Very High (patient harm, massive fines, trust collapse)

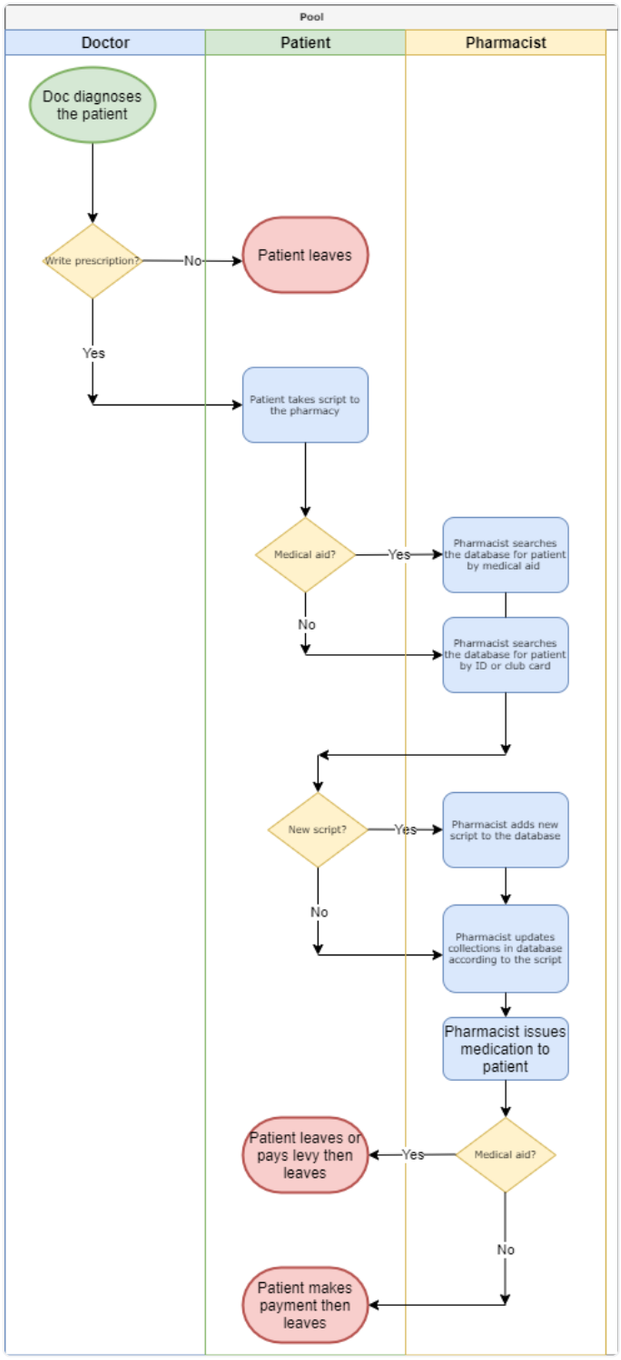
Bottom line: While the app's vision (legal AES e-prescribing + pharmacy loop + adherence tracking + true portable PHR) addresses real needs (illegible scripts, poor adherence, fragmented records), it faces major headwinds — particularly around pharmacist acceptance of patient-forwarded scripts, strict AES/POPIA compliance, building network participation, and cybersecurity threats in a sector already under attack.

Existing compliant platforms succeed by focusing narrowly (doctor-direct transmission, AES-first) and starting in urban/private areas. A full-featured app like this would require regulatory advocacy, strong partnerships (pharmacies, medical aids), phased rollout (lower schedules first), and exceptional security to overcome these issues.If you're considering building or refining this concept, early engagement with SAPC, HPCSA, and the Information Regulator is essential to navigate these challenges.

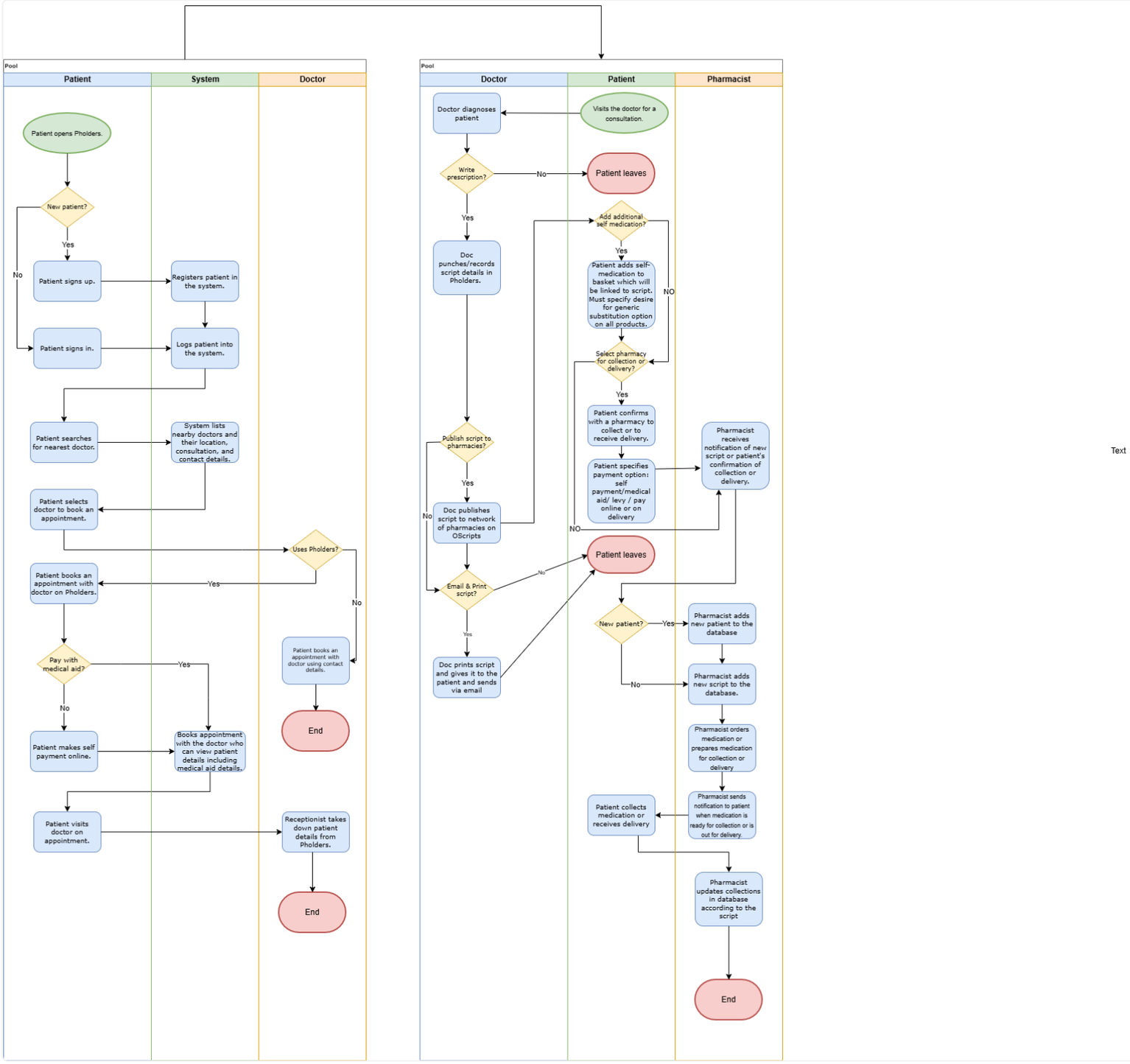
4 Business Process Overview

[Describe how the current process(es) work, including the interactions between systems and various business units. Include visual process flow diagrams to further illustrate the processes the new product will replace or enhance.
Use case documentation and accompanying activity or process flow diagrams can be used to create the description(s) of the proposed or “To-Be” processes.]

4.1 Current Business Process (As-Is)



4.2 Proposed Business Process (To-Be)



Text

5. Business Requirements

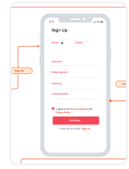
[The specific business requirements elicited from stakeholders should be listed, categorized by both priority and area of functionality to smooth the process of reading and tracking them. Include links to use case documentation, and other key reference material as needed to make the requirements as complete and understandable as possible. You may wish to incorporate the functional and non-functional requirements into a traceability matrix that can be followed throughout the project.]

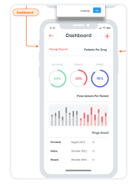
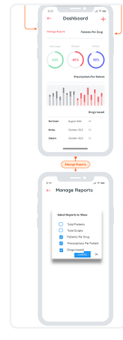
The requirements in this document are prioritized as follows:

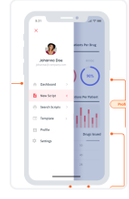
Value	Rating	Description

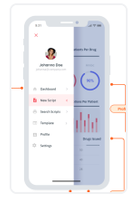
1	Critical	This requirement is critical to the success of the project. The project will not be possible without this requirement.
2	High	This requirement is high priority, but the project can be implemented at a bare minimum without this requirement.
3	Medium	This requirement is somewhat important, as it provides some value but the project can proceed without it.
4	Low	This is a low priority requirement, or a “nice to have” feature, if time and cost allow it.
5	Future	This requirement is out of scope for this project, and has been included here for a possible future release.

5.1 Functional Requirements

Req#	Priority	Description	Rationale	Related Screen	Business Rule Reference	Impacted Stakeholders
General / Base Functionality						
FR-G-001	1	The doctor registration shall include the doctor’s license number, practice number and an indication that he/she is a doctor, and indication of type of doctor. Other registration details required are username and mobile number and password.	This will make the record unique and make sure the doctor is authentic.		BR-035	Development team
FR-G-002	1	An OTP must be sent to verify the doctor’s account.			BR-035	Development team

FR-G-003	1	After the verification of a new doctor’s account the app shall land on the Template screen.			BR-035	Development team
FR-G-004	1	After the verification of an existing doctor’s account the app shall land on the Dashboard.				Development team
FR-G-005	1	The doctor shall supply the rest of his profile information on a template that shows how prescriptions will look.				Development team
FR-G-006	1	The doctor shall supply all his profile information: username, mobile number, email, password, first name, surname, qualification, practice names, practice numbers, logo, profile picture, gender, and signature, consultation rate, and operating hours.				Development team
FR-G-007	1	After login the doctor shall land on the dashboard.				Development team
FR-G-008	1	From the dashboard the doctor shall view the following reports and manage which reports he/she desires to see: Total Patients, Total Scripts, Patients Per Drug, Prescriptions Per Patient, Drugs Issued, and practice-level insights (e.g., patient volume, adherence				Development team

		trends – anonymized/aggregated)				
FR-G-009	1	The doctor shall be able to view his appointments in a calendar.				Development team
FR-G-010	1	The doctor shall create a new prescription for a patient according to the type of doctor he is, through text input or through speech with transcribing AI technology, and capture multiple drugs on that prescription.				Development team
FR-G-011	1	The doctor shall be able to create a new prescription for a patient that is not on the system.				Development team
FR-G-012	1	The system shall send the patient not on the system an sms with a link to download the app.				Development team
FR-G-013	1	The doctor shall view the prescription before publishing it to the network of pharmacies.				Development team
FR-G-014	1	The doctor shall search the system for scripts by patient, drug, medical aid, reference number, from date, to date.				Development team
FR-G-015	1	The doctor shall view, email or print a script after a search.				Development team
FR-G-016	1	A sliding menu shall enable the doctor to easily access the functionality in the system.				Development team

FR-G-017	1	The doctor's sliding menu shall consist of the following items: Dashboard, New Script, Search Scripts, Templates, Calendar, Profile, Settings.				Development team
FR-G-018	2	A doctor shall create, update, or delete sick notes				Development team
FR-G-019	1	A new patient shall be linked to the doctor when the doctor publishes a script.	Each doctor must be linked to their patients for the prescription to be authentic.			Development team
FR-G-020	1	The doctor shall create/update/delete scripts and the patient shall be able to view/print/email his/her own scripts.	Only doctors are authorized to create, update or delete scripts and the patient can view/print/email and cannot change the script.			Development team
FR-G-021	1	Upon registration, the patient shall indicate that he/she is a patient and provide his/her username and mobile number and password.	These are the basic registration details required to simplify the registration process.			Development team
FR-G-022	1	An OTP must be sent to verify the patient’s account.				Development team
FR-G-023	1	If the patient has received a script(s) from a doctor through the system, but the patient doesn’t already exist on the system then the				Development team

		patient shall land on the Search Scripts screen after his/her account has been verified.				
FR-G-024	1	The patient shall be able to view, email or print his/her scripts.				Development team
FR-G-025	1	If the patient has not received a script(s) from a doctor through the system, and the patient doesn't already exist on the system then the patient shall land on a Welcome screen after his/her account has been verified.				Development team
FR-G-026	1	After login the patient shall land on the dashboard.				Development team
FR-G-027	1	From the dashboard the patient shall view the following summary report and manage which variables he/she desires to see: Total Scripts, Nearest Pharmacy, Medical Aid Name, Medical Aid Plan, Medial Aid Number, Last Doctor's Visit, Last Medication Received, Next Collection Date, Last Pharmacy Visited, Favorite Pharmacy.				Development team
FR-G-028	1	The patient shall create a schedule to remind him/her when to take medication and indicate whether the medication has been taken, skipped, or missed.				Development team

FR-G-029	1	The patient shall be able to view details of the doctors he/she is linked to in the system.	For reference.			Development team
FR-G-030	1	The patient shall be able to search and view all nearby doctors.				Development team
FR-G-031	1	The patient shall be able to view a doctor’s location, consultation, and contact details.				Development team
FR-G-032	1	A sliding menu shall enable the patient to easily access the functionality in the system.				Development team
FR-G-033	1	The patient's sliding menu shall consist of the following items: Bookings, Dashboard, Search Scripts, Scheduler, My doctors, Appointments, Profile, Settings.				Development team
FR-G-034	1	The patient shall use a form with access to the selected doctor's calendar to book an appointment with a doctor registered on Pholders.				Development team
FR-G-035	1	The patient shall make a payment for the consultation upon booking an appointment through online self-payment or by providing medical aid details to the doctor.				Development team
FR-G-036	1	The patient shall be able to cancel an appointment				Development team

		in 3 or more working days in advance.				
FR-G-037	1	The patient shall be refunded when a cancellation is made on time.				Development team
FR-G-038	1	The patient shall use a basket to purchase over-the-counter medication.				Development team
FR-G-039	1	The patient's over-the-counter medication shall be linked to a script reference.				Development team
FR-G-040	1	The patient shall specify his/her choice to allow a generic substitution on all his/her selected products including medication on the script.				Development team
FR-G-041	1	The patient shall confirm the collection of medication or delivery of them with a specific pharmacy or go to any pharmacy to order or purchase them.				Development team
FR-G-042	1	The patient shall search for a nearby doctor.				Development team
FR-G-043	1	The patient shall select a doctor and view his/her location, consultation, and contact details.				Development team
FR-G-044	1	If the selected doctor uses Pholders, the patient shall use a form to book an appointment with the doctor.				Development team
FR-G-045	1	If not using a medical aid the patient shall make an online payment before the booking is made.				Development team

FR-G-046	1	If using medical aid the patient’s details including medical aid details shall be sent to the doctor.				Development team
FR-G-047	1	A booking shall be made by sending the booking form and patient details to the doctor and also payment details.				Development team
FR-G-048	1	The pharmacist registration shall include the pharmacist's registration number, pharmacy name, pharmacy practice number, and an indication that he/she is a pharmacist. Other registration details required are username and mobile number and password.	This will make the record unique and make sure the pharmacist is authentic.			Development team
FR-G-049	1	An OTP must be sent to verify the pharmacist’s account.				Development team
FR-G-050	1	After the verification of the pharmacist’s account the app shall land on the Dashboard.				Development team
FR-G-051	1	The pharmacist shall supply the rest of his/her profile information on a profile form.				Development team
FR-G-052	1	The pharmacist shall supply all his/her profile information: username, mobile number, email, password, first name, surname, pharmacist registration number, pharmacy name, pharmacy practice			BR-020	Development team

		number, logo, profile picture, and gender.				
FR-G-053	1	After login the pharmacist shall land on the dashboard.				Development team
FR-G-054	1	On the dashboard the pharmacist shall view all scripts published to the network of pharmacies and be able to claim and print them.				Development team.
FR-G-055	1	When a script is claimed, the pharmacist shall be able to view the script details as well as the additional self-medication, the patient's personal and medical aid details, the patients choice for collection or delivery, the patient's choice for generic substitution on all products, and whether the patient has made a payment, will pay upon delivery, or will pay via medical aid.				Development team
FR-G-056	1	On the dashboard the pharmacist shall be able to mark scripts as processed.				Development team
FR-G-057	1	The system shall notify patients when the pharmacist has viewed, printed or has processed their scripts.				Development team
Security Requirements						
FR-S-001	1	A repository must be used in the development of the app restricting users.	To track the version history and ensure everyone			Development team

			updates the latest version			
FR-S-002	1	Use API's for access control.	To prevent hacking and so that users only get access to what they are allowed.			Development team.
Reporting Requirements						
FR-R-001	1	The app will generate reports.	To track the patient's medical history, drugs received and doses.			Development team
Usability Requirements						
FR-U-001						
Audit Requirements						
FR-A-001	1	Every entry, update, deletion, and unsuccesful attempt by a user to access an item of data shall be audited.	For accountability and tracing errors.			Development Team

5.2 Non-Functional Requirements

[Include technical and operational requirements that are not specific to a function. This typically includes requirements such as processing time, concurrent users, availability, etc.]

ID	Requirement
NFR-001	The system shall handle a million users without affecting it's performance.
NFR-002	Every update, deletion and unsuccessful attempt by a user to access an item of data shall be recorded on an audit trail.
NFR-003	The system shall work on multiple operating systems: Android 6.0 and up, iOS 15.0 and up, Windows 10.0 and up, and macOS 10.14 and up.
NFR-004	The system shall work on all screen resolutions.
NFR-005	The system shall use Postgres SQL for a database.

NFR-006	Flutter shall be used for prototyping.
NFR-007	

6. Business Rules

[A business rule defines or constrains some aspect of business and always resolves to either true or false. For example, a business rule might state that no credit check is to be performed on return customers. Other examples of business rules include requiring a rental agent to disallow a rental tenant if their credit rating is too low, or requiring company agents to use a list of preferred suppliers and supply schedules.]

ID	Name	Description	Related Screen
BR-001	Medical aid	If the patient is on medical aid, the medical aid name, medical aid number, medical aid plan, and/or dependent number (s) must be filled in the profile form.	n/a
BR-002	Medical aid	If the patient is on medical aid, the medical aid name, medical aid number, and/or dependent number must also be sent to the pharmacy.	n/a
BR-003	Medical Aid	B2B partnerships with medical aids must be established as they pay to integrate for better chronic care management, formulary checks, cost savings, and member engagement.	n/a
BR-004	Medical Aid - Patient Verification & Medical Aid Linkage	Mandatory member validation — Patient must verify medical aid membership (e.g., membership number, ID number, scheme name) via secure input or integration. App must cross-check against scheme databases (where API access exists) to confirm active status, dependants, and primary beneficiary.	n/a
BR-005	Medical Aid - Patient Verification & Medical Aid Linkage	Consent for data access — Explicit, granular POPIA-compliant consent required before querying any medical aid switch for benefit checks, claims status, or formulary info. Consent must be revocable and logged.	n/a
BR-006	Medical Aid - Patient Verification & Medical Aid Linkage	No automatic enrollment/registration — App cannot register patients on chronic programs or submit pre-authorizations on behalf of schemes unless explicitly partnered and	n/a

		authorized (e.g., via Discovery HealthID protocols).	
BR-007	Medical Aid - Real-Time Benefit & Formulary Check (Critical for Prescribing)	Real-time eligibility & benefit verification (RTBV/RTPB) — When doctor creates prescription, app must (if integrated) query the relevant switch (e.g., Medscheme, Medikredit) to display: • Whether medicine is covered (acute/chronic benefit). • Co-payment amount (if any). • Formulary status (preferred/generic vs. non-formulary). • Remaining benefit limits (e.g., annual chronic medication allocation).	n/a
BR-008	Medical Aid - Real-Time Benefit & Formulary Check (Critical for Prescribing)	Formulary enforcement — Highlight non-formulary items with warnings; suggest generics/alternatives where scheme incentivizes (e.g., via lower co-pays). Doctor can override with clinical motivation, but app must log justification.	n/a
BR-009	Medical Aid - Real-Time Benefit & Formulary Check (Critical for Prescribing)	PMB compliance — For Prescribed Minimum Benefits (25 chronic conditions + emergencies), app must flag PMB-eligible items and ensure coverage at cost (or scheme rate) without co-pays, per CMS rules.	n/a
BR-010	Medical Aid - Real-Time Benefit & Formulary Check (Critical for Prescribing)	No misleading cost displays — Any pricing/estimate shown must be real-time from switch (not static) and include disclaimers (e.g., "subject to final scheme adjudication").	n/a
BR-011	Medical Aid - Prescription & Dispensing Integration	NAPPI code mandatory — Every prescribed item must use valid NAPPI codes (9-digit, product + pack size) for accurate scheme adjudication and reimbursement.	n/a
BR-012	Medical Aid - Prescription & Dispensing Integration	Chronic medication rules — For registered chronic patients, app must support repeat dispensing (up to scheme limits, e.g., 3–6 months) with auto-refill reminders, but only	n/a

		if patient is enrolled on scheme chronic program.	
BR-013	Medical Aid - Prescription & Dispensing Integration	Pharmacy network alignment — When patient sends script to network pharmacy, app must prioritize DSPs (Designated Service Providers) if scheme mandates (e.g., Discovery MedXpress or scheme-preferred pharmacies) to avoid reduced benefits/co-pays.	n/a
BR-014	Medical Aid - Prescription & Dispensing Integration	No automatic claims submission — App can facilitate pre-adjudication visibility but cannot submit claims directly unless B2B agreement exists (most schemes require pharmacy/doctor submission).	n/a
BR-015	Medical Aid - Adherence Tracking & Reporting	Adherence data sharing restricted — Patient adherence logs (intake tracking, reminders) can be used for personal PHR but sharing with schemes (e.g., for wellness incentives or chronic program compliance) requires explicit opt-in consent.	n/a
BR-016	Medical Aid - Adherence Tracking & Reporting	Aggregated/anonymized insights only — Any reports to schemes (if partnered) must be anonymized/aggregated to avoid POPIA breaches.	n/a
BR-017	Medical Aid - Adherence Tracking & Reporting	No penalty enforcement — App cannot withhold features or penalize non-adherence; schemes may use data for interventions (e.g., counseling), but app provider has no authority to enforce.	n/a
BR-018	Medical Aid - Partnership & Integration	Scheme-specific APIs/partnerships required — Full real-time checks need formal agreements with switches (e.g., Discovery HealthID for e-scripting + benefit visibility, or Scrypt-O/PrescriBIT-style APIs). Without integration, features degrade to manual input/estimates.	n/a
BR-019	Medical Aid - Partnership & Integration	CMS oversight compliance — App must not interfere with scheme governance (e.g., no bypassing PMB rules, no inducing over-utilization). Features like doctor advertising or ratings must not violate HPCSA/CMS advertising guidelines.	n/a

BR-020	Medical Aid - Partnership & Integration	Data residency & security — All medical aid-related data processing must comply with POPIA + scheme contracts (often requiring SA-based cloud storage).	n/a
BR-021	Pharmacies	B2B partnerships with a network of pharmacies must be established for agreements with chains/independents to claim prescriptions + handle orders/delivery.	n/a
BR-022	Pharmacies - Prescription Transmission & Authentication (Highest Priority)	AES mandatory for legal validity — Every digital prescription must be signed using a valid Advanced Electronic Signature (AES) from an accredited provider (e.g., TrustFactory-integrated with OTP/2FA). The app must enforce this before finalization/transmission; unsigned or non-AES scripts are invalid and cannot be sent/claimed.	n/a
BR-023	Pharmacies - Prescription Transmission & Authentication (Highest Priority)	Pharmacist must verify authenticity — Per Regulation 33(4), the dispensing pharmacist is responsible for verifying the prescription's authenticity before dispensing. The app must provide clear, visible indicators (e.g., TrustFactory seal, AES verification badge, QR code linking to audit trail) on transmitted scripts/PDFs for easy pharmacist check.	n/a
BR-024	Pharmacies - Prescription Transmission & Authentication (Highest Priority)	Direct transmission strongly preferred — SAPC guidance encourages direct prescriber-to-pharmacy transmission to minimize authentication risks. Patient-mediated transmission (patient receives script in app → forwards/claims to pharmacy) is permissible but places extra burden on pharmacists → potential delays/refusals, especially for S5/S6. The app should log/warn users about this and prioritize direct routing where possible (e.g., patient selects pharmacy → app routes directly).	n/a
BR-025	Pharmacies - Prescription Transmission & Authentication (Highest Priority)	Immutable audit trail required — Full tracking history of script actions (creation, signing, transmission, claims, dispensing status) must be stored securely and visible to pharmacists/doctors (e.g., via pharmacy	n/a

		portal). This supports fraud prevention and legal defense.	
BR-026	Pharmacies - Pharmacy Network & Claiming	Single pharmacy claim lock — Once a pharmacy claims a prescription (via app network), it must lock immediately to prevent duplicate dispensing/claims by others. Real-time status updates (pending → claimed → preparing → ready/collected/delivered) must be pushed to patient and prescriber.	n/a
BR-027	Pharmacies - Pharmacy Network & Claiming	Network pharmacy requirements — Participating pharmacies must be: • Registered with SAPC and licensed by Department of Health. • Compliant with GPP (responsible pharmacist oversight, secure storage/delivery). • Able to verify AES/authenticity. • For online purchase/delivery: Must operate as a licensed pharmacy (website must display Y-number, responsible pharmacist name, consultation hours); no dispensing without pharmacist review.	n/a
BR-028	Pharmacies - Pharmacy Network & Claiming	Stock visibility & preparation — If app shows real-time stock (optional), it must come from pharmacy integrations (not static data). Pharmacy must prepare order only after claim + verification; patient notified (push/SMS) when ready for collection/delivery.	n/a
BR-029	Pharmacies - Dispensing & Delivery	Pharmacist oversight mandatory — No automatic dispensing; responsible pharmacist must review/verify script, counsel patient (in-person or via consultation for delivery), and ensure safe/effective use (GPP 2.7).	n/a
BR-030	Pharmacies - Dispensing & Delivery	Delivery rules — For online purchase/delivery: • Secure logistics (cold chain for thermolabile items, signature on delivery).	n/a

		• Audit trail for delivery process. • No bypassing pharmacist consultation (SAPC rules for distance selling).	
BR-031	Pharmacies - Dispensing & Delivery	Higher-schedule restrictions — S5/S6 scripts face stricter verification; app should flag potential issues and encourage direct transmission or in-person pickup where needed.	n/a
BR-032	Pharmacies - Integration & Technical	API/secure channel integration — Pharmacy systems (e.g., HealthWare, ProPharm, or emerging platforms) must receive scripts securely (encrypted, auditable). Many current platforms use APIs for push notifications, claim locking, and status sync.	n/a
BR-033	Pharmacies - Integration & Technical	No bypassing regulations — App cannot allow dispensing without pharmacist intervention or facilitate illegal supply (e.g., no prescription meds without valid script).	n/a
BR-034	Pharmacies - Integration & Technical	Record-keeping — Pharmacies must retain electronic scripts as per GPP (patient medication records, prescription book); app should support export/print for compliance.	n/a
BR-035	Doctor license (registration) number and practice number	The doctor must provide his/her license and practice numbers upon registration which must be verified by an API.	n/a
BR-036	Email addresses	Entered email addresses must appear valid (contain “@” and “.”)	n/a
BR-037	Mobile numbers	Entered mobile numbers must appear valid and formatted.	n/a
BR-038	Passwords	Entered passwords must be strong and encrypted.	n/a
BR-039	Doctor and patient relationship	Doctors must have one or more patients and patients can have one or more doctors.	n/a
BR-040	Pop-ups	Confirmation, error, or info messages must be formatted the same way and pop up in a message dialog box.	n/a
BR-041	SMS	Sms’s must be short and concise.	n/a

BR-042	Script Archives	After 6 months, the scripts must be unprintable and not downloadable. Only viewable. Because some pharmacies stop archiving them after 6 months by law.	n/a
BR-043	Doctors - Prescribing & E-Signature (Core – Highest Priority)	AES required for all prescriptions — Doctors must apply a valid Advanced Electronic Signature (accredited provider, OTP/2FA verification) to finalize any digital prescription. The app must block issuance without successful AES. Unsigned or non-AES scripts are invalid and cannot be transmitted/shared.	n/a
BR-044	Doctors - Prescribing & E-Signature (Core – Highest Priority)	Established practitioner-patient relationship preferred — Prescribing via the app should follow an established relationship (in-person or prior telehealth). While not strictly compulsory in all cases (per updated Booklet 10 flexibility post-COVID), first-time prescribing based solely on questionnaires/forms/online data is not an acceptable standard of care and must be prohibited or heavily flagged.	n/a
BR-045	Doctors - Prescribing & E-Signature (Core – Highest Priority)	Proper assessment before prescribing — Doctors must ensure adequate history/examination (via prior consultation or app-enabled telehealth if compliant). The app cannot allow diagnosis/treatment initiation purely from patient-input data without doctor judgment.	n/a
BR-046	Doctors - Prescribing & E-Signature (Core – Highest Priority)	Mandatory fields & NAPPI codes — Prescriptions must include: patient details (ID/name), doctor HPCSA number/contact, medicine name/strength/dosage/instructions/quantity, date, schedule. Use valid NAPPI codes for medicines to support medical aid adjudication.	n/a
BR-047	Doctors - Prescribing & E-Signature (Core – Highest Priority)	No automatic/over-prescribing — Avoid perverse incentives; prescriptions must be clinically justified. App must log clinical rationale if overriding formulary suggestions.	n/a

BR-048	Doctors - Telehealth & Consultation (If App Supports Remote Elements)	Telehealth only when appropriate — Use for follow-ups, chronic management, or where clinically suitable. Initial consultations should ideally be in-person unless justified (Booklet 10 allows flexibility but emphasizes standard of care).	n/a
BR-049	Doctors - Telehealth & Consultation (If App Supports Remote Elements)	Informed consent for telehealth — Obtain and document patient consent for using telehealth (if applicable) and for electronic prescribing/transmission.	n/a
BR-050	Doctors - Telehealth & Consultation (If App Supports Remote Elements)	Maintain professional standards — Same ethical duties apply (confidentiality, autonomy, non-maleficence). No advice/treatment without proper assessment.	n/a
BR-051	Doctors - Patient Record & Data Handling	Accurate, secure record-keeping — Doctors must ensure app-generated prescriptions/records are accurate and integrated into their own patient files where required (HPCSA Booklet 9 on Patient Records). The portable PHR is patient-owned, but doctors remain responsible for their contributions.	n/a
BR-052	Doctors - Patient Record & Data Handling	Confidentiality & POPIA compliance — Only access patient data with consent; report breaches promptly. Doctors must not misuse shared PHR data (e.g., for marketing without consent).	n/a
BR-053	Doctors - Patient Record & Data Handling	Patient consent for sharing — When viewing patient-uploaded evidence (e.g., photos of generics) or adherence logs, doctors must obtain/use only with explicit permission.	n/a
BR-054	Doctors - Advertising & Profile	Advertising must be truthful & professional — Doctors can advertise practice details (location, specialties, hours, contact) via the app but cannot make unsubstantiated claims (e.g., "best doctor," "guaranteed results"), offer inducements, discounts unrelated to care, or tout services (HPCSA Booklet 16 on Social Media & advertising rules).	n/a

BR-055	Doctors - Advertising & Profile	No canvassing/touting — Cannot exchange services for reviews, publicity, or payments; no misleading comparisons.	n/a
BR-056	Doctors - Advertising & Profile	Ratings response allowed — Doctors can respond to patient reviews professionally but cannot solicit fake ratings or retaliate.	n/a
BR-057	Doctors - Booking, Reminders & Practice Management	Availability accuracy — Doctors must keep calendars accurate; overbooking or no-shows damage professional reputation.	n/a
BR-058	Doctors - Booking, Reminders & Practice Management	Appointment reminders — Doctors should honor scheduled bookings; use app reminders to reduce no-shows but remain responsible for follow-up.	n/a
BR-059	Doctors - Booking, Reminders & Practice Management	Reports usage — Access to generated reports (e.g., adherence trends, patient volume) must be for legitimate practice improvement only; no unauthorized sharing.	n/a
BR-060	Doctors - General Professional Conduct	HPCSA registration verification — Doctors must maintain active HPCSA registration; app must verify before onboarding/continued use.	n/a
BR-061	Doctors - General Professional Conduct	No conflicts of interest — Avoid promoting specific pharmacies/products unless clinically justified.	n/a
BR-062	Doctors - General Professional Conduct	Liability awareness — Doctors remain professionally liable for prescriptions issued via the app; app disclaimers should reinforce this.	n/a
BR-063	Patients - Consent & Data Privacy (POPIA – Core for Health Data)	Provide explicit, informed, voluntary consent — Before any processing of special personal information (e.g., medical aid details, prescriptions, adherence logs, uploaded photos), patients must give specific, granular consent (e.g., separate checkboxes for sharing with pharmacies, doctors viewing PHR, reminders). Consent must be withdrawable at any time.	n/a

BR-064	Patients - Consent & Data Privacy (POPIA – Core for Health Data)	Provide accurate & complete personal information — Patients must supply truthful details (ID, medical aid membership, allergies, conditions) when registering/linking profiles or during onboarding. False information invalidates features and may lead to account suspension.	n/a
BR-065	Patients - Consent & Data Privacy (POPIA – Core for Health Data)	Notify app of changes — Patients must update personal/medical details (e.g., new medical aid, changed allergies) promptly to ensure accurate records and safe prescribing.	n/a
BR-066	Patients - Consent & Data Privacy (POPIA – Core for Health Data)	Exercise data subject rights responsibly — Patients can access, correct, delete, or object to processing of their data (POPIA Sections 23–25), but repeated frivolous requests may be limited per law.	n/a
BR-067	Patients - Consent & Data Privacy (POPIA – Core for Health Data)	Report suspected breaches — Patients must notify the app immediately of any unauthorized access or suspected data compromise.	n/a
BR-068	Patients - Prescription & Medication Use	Do not alter, duplicate, or misuse digital prescriptions — Patients must not edit, forge, share for unauthorized use, or attempt multiple claims/dispensings of the same script. Violations can lead to account termination and potential legal reporting (fraud under Medicines Act).	n/a
BR-069	Patients - Prescription & Medication Use	Use prescriptions lawfully — Send/claim scripts only for personal use or authorized dependants. Present valid ID when collecting/delivery if required by pharmacy.	n/a
BR-070	Patients - Prescription & Medication Use	Accurately log medication intake — For adherence tracking/reminders to be reliable, patients should log intake truthfully (taken/skipped/missed). Inaccurate logging undermines personal health management and shared reports (if consented).	n/a
BR-071	Patients - Prescription &	Upload genuine evidence only — Photos of generics/alternatives or pill bottles must be	n/a

	Medication Use	real and relevant to own records; no misleading uploads.	
BR-072	Patients - Booking, Appointments & Interaction	Provide accurate booking information — Select correct timeslots, provide truthful reasons/symptoms (where requested), and cancel/reschedule promptly to avoid wasting doctor time.	n/a
BR-073	Patients - Booking, Appointments & Interaction	Honor appointments — Attend booked consultations (in-person or telehealth if applicable); repeated no-shows may lead to restrictions on future bookings.	n/a
BR-074	Patients - Booking, Appointments & Interaction	Give honest feedback in ratings/reviews — Reviews must be fair, factual, and post-consultation; no fake reviews, harassment, or inducements (e.g., for better ratings).	n/a
BR-075	Patients - Pharmacy & Purchase	Select legitimate network pharmacies — Only use registered SAPC pharmacies in the app network for claiming/ordering. Report suspicious pharmacies.	n/a
BR-076	Patients - Pharmacy & Purchase	Follow pharmacist instructions — For collection/delivery: provide required verification, accept counseling, and adhere to delivery terms (e.g., signature, cold chain).	n/a
BR-077	Patients - Pharmacy & Purchase	Pay for services fairly — Settle any co-payments, delivery fees, or online purchases as required; no chargebacks without valid reason.	n/a
BR-078	Patients - General Conduct & PHR Management	Treat the portable PHR responsibly — Use the lifelong record for legitimate healthcare continuity (sharing with providers with consent). Do not share login credentials or allow unauthorized access.	n/a
BR-079	Patients - General Conduct & PHR Management	Age & guardian rules — Patients under legal age must involve guardians/parents for consent/actions (per POPIA/National Health Act); app enforces age gates.	n/a
BR-080	Patients - General Conduct & PHR Management	No abusive/misuse of features — No spam bookings, fake reviews, harassment of doctors, or using app for illegal purposes (e.g., sourcing controlled substances fraudulently).	n/a

BR-081	Appointment Booking	The system must allow a patient to book an appointment with a doctor who uses it.	n/a
BR-082	Appointment Booking	Payments for consultations must be made beforehand upon booking an appointment.	n/a
BR-083	APIs	The system must be able to integrate with existing healthcare systems or electronic medical record (EMR) systems to send patient data when the patient is at the doctor’s reception.	n/a
BR-084	APIs	The system must be able to integrate with the doctor’s appointment schedule to make non conflicting appointments.	n/a
BR-085	APIs	The system must be able to integrate with a pharmacy’s system to send scripts and patient data.	n/a
BR-086	Payment Process	The patient must make purchases online as self-payment, or offline upon delivery, or specify medical aid details to charge his medical insurance.	n/a
BR-087	Payment Process	The patient must be able to pay a levy online or upon delivery.	n/a
BR-088	Payment Process	The patient can purchase over-the-counter medication which purchase will be linked by a script reference.	n/a
BR-089	Payment Process	The patient must specify if he/she wants a generic substitution on all products.	n/a
BR-090	Logins	The doctor, patient, and pharmacist must be able to login into the system and supply profile information.	n/a
BR-091	Nearby doctor	The patient must be able to find a doctor nearest to them and if under medical aid, a doctor that’s covered by his/her medical plan.	n/a
BR-092	Ratings and Review	The system must allow the patient to rate and review a doctor’s service.	n/a
BR-093	Notifications and reminders	The system must notify the patient to remind him/her of appointments.	n/a
BR-094	Deliveries	The system must incorporate a courier delivery service for the delivery of	n/a

		medication to patients.	
BR-095	Notifications and reminders	The system must get updates from pharmacy of scripts and generics that need to be updated.	n/a
BR-096	Uploads	The system must allow the patient to upload pictures of generics or alternative medicine for his/her own records and evidence.	n/a
BR-097	Medical Records	The system must allow any doctor that is signed up to access his/her patient's medical records.	n/a
BR-098	Schedule Medication Intake	The system must allow the patient to save medication and take pictures of medication and schedule when to use them.	n/a
BR-099	E-Prescribing & Prescription Validity Rules (Core – Highest Priority)	All prescriptions must be signed using a valid Advanced Electronic Signature (AES) — Doctor must use an accredited AES (e.g., via OTP/2FA from providers like TrustFactory) for all schedules (including S5/S6). No script can be issued/finalized without AES verification. Simple digital signatures, PDFs without AES, or wet-ink scans are invalid.	n/a
BR-100	E-Prescribing & Prescription Validity Rules (Core – Highest Priority)	Prescription must be generated by an "electronic agent" compliant with ECTA + Regulation 33 — The app acts as the electronic agent; it must enforce mandatory fields (patient ID/name, prescriber details/HPCSA number, medicine name/strength/dosage/instructions/quantity, date, schedule classification).	n/a
BR-101	E-Prescribing & Prescription Validity Rules (Core – Highest Priority)	Pharmacist authentication required — Every digital prescription transmitted (even via patient) must include visible AES indicators/seals/QR for pharmacist verification of authenticity before dispensing (Regulation 33(4)).	n/a
BR-102	E-Prescribing & Prescription Validity Rules (Core – Highest Priority)	No first-time prescribing based solely on app data/questionnaires — Prescribing only allowed after an established doctor-patient relationship and proper assessment (HPCSA Telehealth Guidelines). App must block or flag any attempt at diagnosis/treatment	n/a

		initiation via forms alone — not an acceptable standard of care.	
BR-103	E-Prescribing & Prescription Validity Rules (Core – Highest Priority)	Direct transmission preferred; patient-mediated allowed but restricted — SAPC strongly recommends doctor-to-pharmacy direct send. Patient-forwarded scripts (via app link/QR) are permissible but trigger extra pharmacist validation steps — app must log/warn about potential delays/refusals, especially for higher schedules.	n/a
BR-104	E-Prescribing & Prescription Validity Rules (Core – Highest Priority)	Validity periods & repeats enforced — Scripts expire (e.g., 30 days for many; no repeats for certain schedules unless specified). App must prevent dispensing beyond legal limits and block unauthorized repeats.	n/a
BR-105	Data Protection & Privacy Rules (POPIA – Special Personal Information)	Health data processing only with explicit, granular patient consent — Consent must be specific, informed, freely given, and withdrawable. App must record/store consent proofs and allow easy withdrawal (deletion of records).	n/a
BR-106	Data Protection & Privacy Rules (POPIA – Special Personal Information)	Patient ownership & control of PHR — Patient must have full access, export (PDF/print/email), correction, and deletion rights. App cannot withhold records or charge excessive fees for access.	n/a
BR-107	Data Protection & Privacy Rules (POPIA – Special Personal Information)	Strict minimization & purpose limitation — Collect/store only necessary data (e.g., no unnecessary profiling). Use only for intended purposes (e.g., adherence tracking, sharing with providers/pharmacies with consent).	n/a
BR-108	Data Protection & Privacy Rules (POPIA – Special Personal Information)	Encryption & security mandatory — End-to-end encryption at rest/transit; immutable audit logs for every access/action; regular security assessments. Breach notification to Information Regulator & affected patients within required timelines.	n/a
BR-109	Data Protection & Privacy Rules (POPIA – Special Personal Information)	No cross-border transfers without safeguards — Cloud/data storage must comply with POPIA adequacy (preferably SA/Africa region; e.g., AWS Cape Town).	n/a

BR-110	Pharmacy Network & Dispensing Rules	Single pharmacy claim only — Once a pharmacy claims a prescription, it locks for others to prevent duplicate dispensing. App must enforce real-time claim status.	n/a
BR-111	Pharmacy Network & Dispensing Rules	Online purchase/delivery restricted — Only registered pharmacies can fulfill; must comply with SAPC rules for distance selling/delivery of medicines (no automatic dispensing without pharmacist oversight).	n/a
BR-112	Pharmacy Network & Dispensing Rules	Stock & formulary visibility optional but compliant — If shown, must be accurate/real-time via pharmacy integrations; no misleading availability claims.	n/a
BR-113	User & Access Control Rules	Doctor verification required — HPCSA registration check before onboarding; AES certificate linkage mandatory.	n/a
BR-114	User & Access Control Rules	Patient age & consent rules — Minors (under 12/18 depending on context) require guardian involvement; app must enforce age-appropriate access.	n/a
BR-115	User & Access Control Rules	No clinical decision support that replaces judgment — Basic interaction alerts ok, but no automated prescribing/dosage suggestions that could be seen as AI-driven care (risk of SAHPRA medical device classification).	n/a
BR-116	Reporting, Ratings & Advertising Rules	Doctor advertising compliant — Ads must not make unsubstantiated claims; follow HPCSA advertising rules (no inducements, truthful).	n/a
BR-117	Reporting, Ratings & Advertising Rules	Ratings moderated — Reviews must be post-consultation; app must prevent fake/manipulated ratings and allow doctor response.	n/a
BR-118	Reporting, Ratings & Advertising Rules	Reports anonymized where required — Doctor reports (e.g., adherence trends) must use aggregated/anonymized data unless explicit patient consent for identifiable use.	n/a
BR-119	System & Platform	Immutable audit trail for all critical actions — Every prescription creation, AES signing,	n/a

	Integrity	transmission, claim, intake log, PHR access/share, upload, rating, and report generation must be logged with timestamp, user ID, IP (where applicable), and action details. Logs must be tamper-proof, retained for at least 5–7 years (aligning with healthcare record retention norms), and accessible for regulatory audits or disputes.	
BR-120	System & Platform Integrity	Version control & change history for prescriptions/PHR — Once AES-signed, prescriptions cannot be edited (only annotated/cancelled by doctor with justification). PHR entries (e.g., uploaded photos, adherence logs) must track versions/changes with timestamps and reasons.	n/a
BR-121	System & Platform Integrity	Single active prescription per claim — Enforce one-time claiming per script instance; prevent reuse or parallel claims across pharmacies.	n/a
BR-122	System & Platform Integrity	No offline finalization of high-risk actions — AES signing, prescription transmission, and pharmacy claiming must require online connectivity + real-time verification to prevent fraud/duplication.	n/a
BR-123	Security & Data Protection (POPIA + Cybersecurity Focus)	End-to-end encryption & data residency — All health data (prescriptions, PHR, medical aid details, photos, adherence logs) encrypted at rest (AES-256+) and in transit. Data storage must be in South Africa or equivalent-protection jurisdictions (POPIA adequacy requirement; prefer AWS Cape Town region).	n/a
BR-124	Security & Data Protection (POPIA + Cybersecurity Focus)	Mandatory breach notification protocol — Detect and notify Information Regulator + affected users within 72 hours (or sooner per POPIA). App must have incident response plan, regular pen-testing, and vulnerability management.	n/a
BR-125	Security & Data Protection (POPIA + Cybersecurity Focus)	Role-based access control (RBAC) — Enforce least-privilege: patients control their PHR sharing; doctors see only consented records; pharmacies see only claimed scripts; no cross-user access without explicit consent.	n/a

BR-126	Security & Data Protection (POPIA + Cybersecurity Focus)	Multi-factor authentication (MFA) — Required for all logins (especially doctors for AES flows and patients linking medical aid).	n/a
BR-127	Security & Data Protection (POPIA + Cybersecurity Focus)	No monetization of identifiable health data — Anonymized/aggregated insights only for reports/research; no sale of personal health information without explicit opt-in.	n/a
BR-128	SAHPRA & Medical Device Classification (Emerging 2025–2026)	Self-assess for SaMD risk — The app (as software intended for diagnosis/monitoring/treatment support via adherence tracking, interaction alerts, or PHR continuity) may qualify as Software as a Medical Device (SaMD). Apply SAHPRA classification rules (Class A–D); higher-risk features (e.g., influencing clinical decisions) likely Class C/D → mandatory SAHPRA registration/licensing if determined applicable.	n/a
BR-129	SAHPRA & Medical Device Classification (Emerging 2025–2026)	No unregistered medical device claims — Avoid marketing features as diagnostic/therapeutic unless registered. Basic adherence reminders/tracking typically low-risk (Class A/B), but advanced analytics or AI elements trigger higher scrutiny.	n/a
BR-130	SAHPRA & Medical Device Classification (Emerging 2025–2026)	Post-market surveillance — Monitor adverse events (e.g., misuse leading to non-adherence harm) and report to SAHPRA if classified.	n/a
BR-131	Operational & Anti-Abuse	Rate limiting & anti-fraud controls — Limit bookings per user/doctor to prevent abuse; flag suspicious patterns (e.g., rapid multiple pharmacy claims, bulk uploads).	n/a
BR-132	Operational & Anti-Abuse	Content moderation for ratings/reviews — Reviews must be post-consultation, factual, non-defamatory; app moderates/removes violations (HPCSA advertising rules apply indirectly).	n/a

BR-133	Operational & Anti-Abuse	Advertising compliance — Doctor ads must follow HPCSA rules (truthful, no inducements); platform cannot guarantee visibility or outcomes.	n/a
BR-134	Operational & Anti-Abuse	Age & capacity gating — Minors require guardian consent; app must enforce age verification and proxy access rules.	n/a
BR-135	Operational & Anti-Abuse	Dispute & escalation handling — Provide in-app mechanisms for reporting issues (e.g., invalid script, data error); escalate to HPCSA/SAPC/SAHPRA/Information Regulator where required.	n/a
BR-136	Reporting & Transparency	Annual compliance reporting — App operator must maintain records for audits by SAHPRA, SAPC, HPCSA, CMS, or Information Regulator (e.g., consent logs, breach incidents).	n/a
BR-137	Reporting & Transparency	Transparency on data use — Publish privacy policy, terms, and data processing details; inform users of any AI/ML elements (per SAHPRA 2025 guidance).	n/a
BR-138	Reporting & Transparency	No interference with regulatory reporting — Facilitate (but not automate without consent) adverse event reporting to SAHPRA or scheme chronic program compliance.	n/a
BR-139	Patients - Personal Health Record (PHR) Categories	The PHR must be categorized by personal details, medical details, financial records, and legal and estate documents, with additional fields and the ability to add custom fields.	n/a
BR-140	Patients - Personal Health Record (PHR) Financial Records (Estate and Asset Management)	The financial record will track the patient's investments and policies which a trusted family member or friend can access to make claims upon the patient's passing.	n/a

7. Appendices

7.1 List of Acronyms

[If needed, create a list of acronyms used throughout the SRS document to aid in comprehension.]

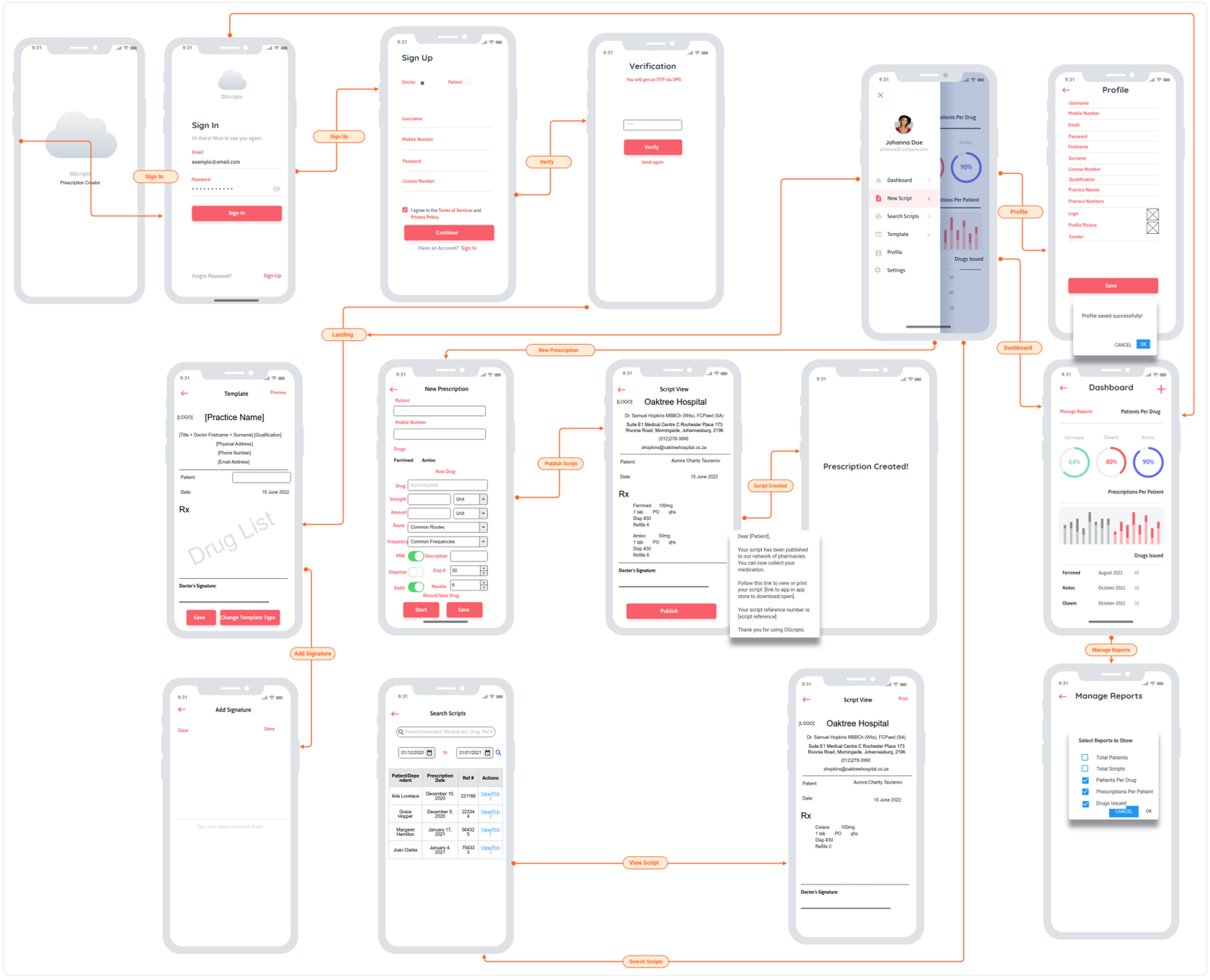
7.2 Glossary of Terms

[If needed, identify and define any terms that may be unfamiliar to readers, including terms that are unique to the organization, the technology to be employed, or the standards in use.]

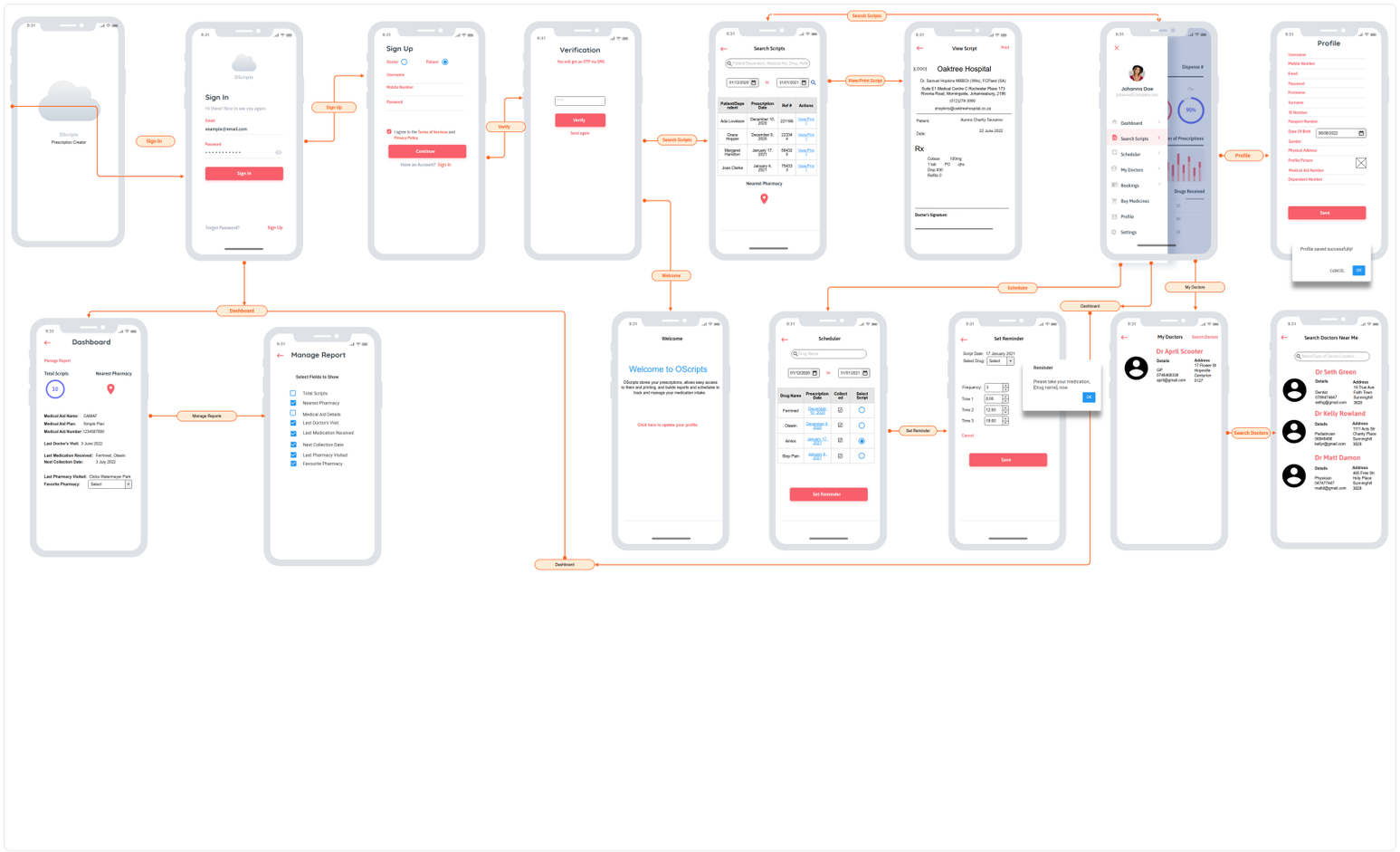
7.3 Related Documents

[Provide a list of documents or web pages, including links, which are referenced in the SRS.]

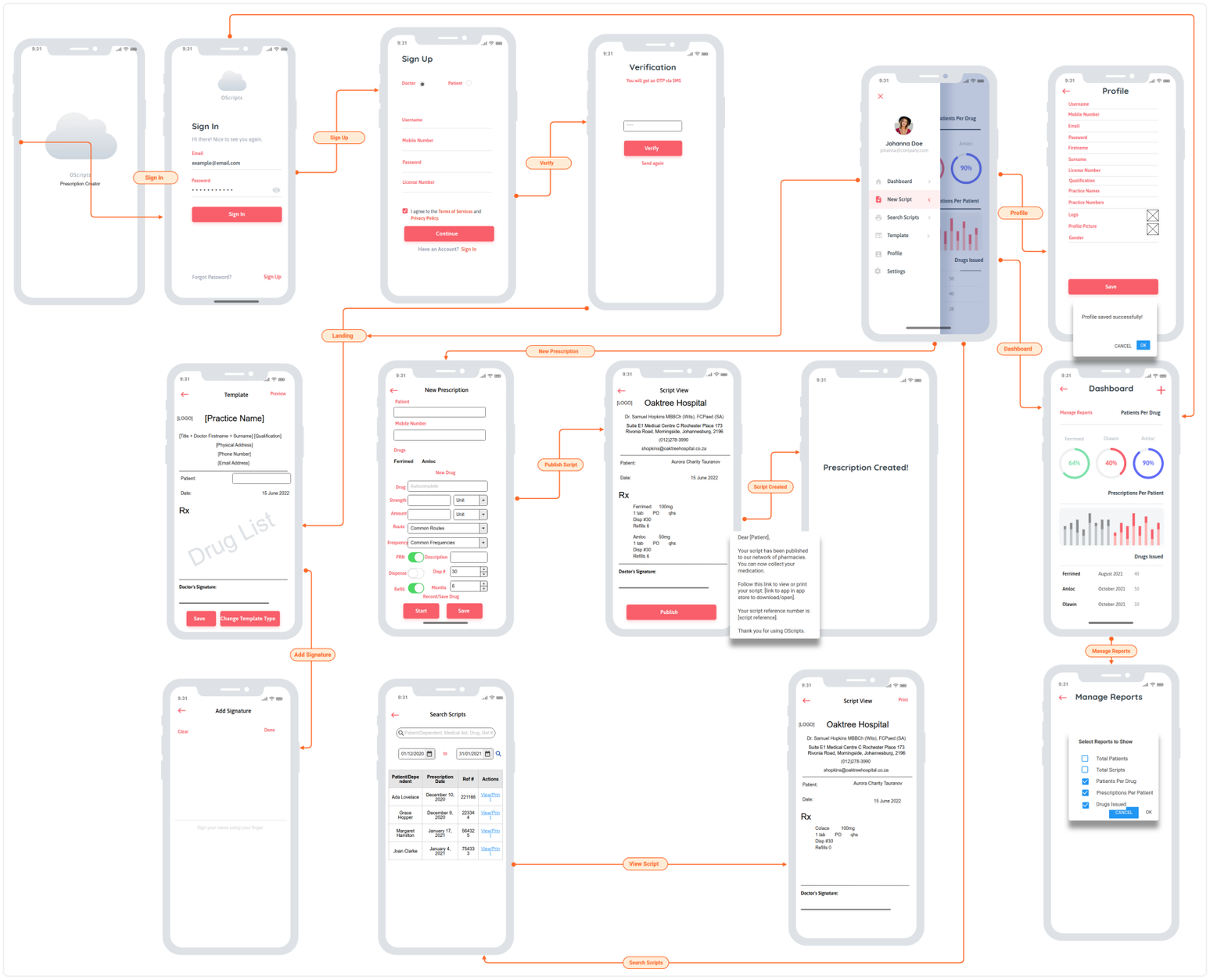
PATIENT BOOKING



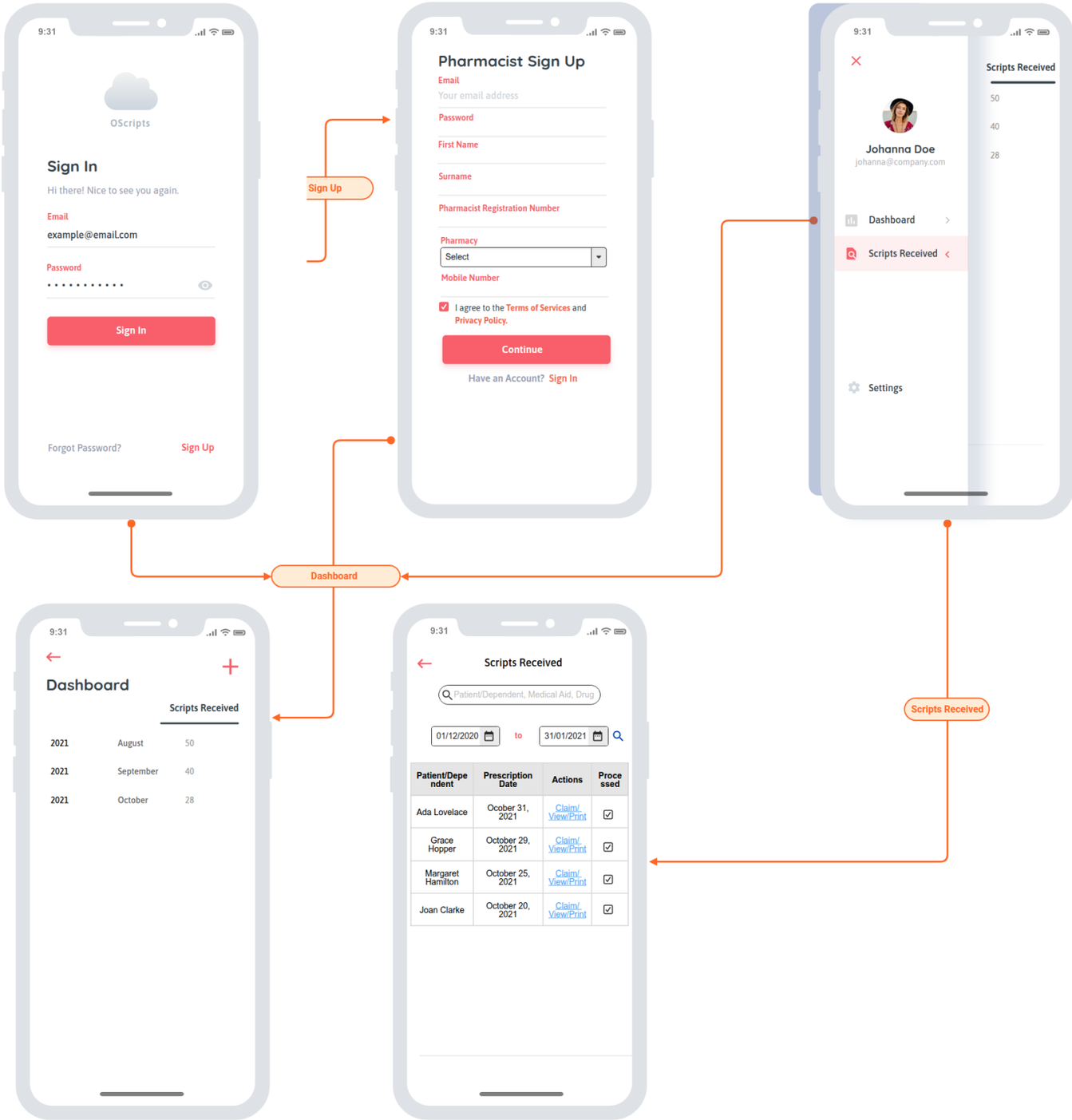
PATIENT SIGN IN



DOCTOR SIGN IN



PHARMACY SIGN IN


9:31
OScripts
Sign In
Hi there! Nice to see you again.
Email
example@email.com
Password
Sign In
Forgot Password?
Sign Up
Sign Up
9:31
Pharmacist Sign Up
Email
Your email address
Password
First Name
Surname
Pharmacist Registration Number
Pharmacy
Select
Mobile Number
I agree to the Terms of Services and Privacy Policy.
Continue
Have an Account? Sign In
9:31
Scripts Received
50
40
28
Johanna Doe
johanna@company.com
Dashboard
Scripts Received
Settings
Dashboard
9:31
Dashboard
Scripts Received
2021 August 50
2021 September 40
2021 October 28
9:31
Scripts Received
Patient/Dependent, Medical Aid, Drug
01/12/2020
to
31/01/2021
Patient/Dependent
Prescription Date
Actions
Processed
Ada Lovelace
Ocober 31, 2021
Claim/ View/Print
Grace Hopper
October 29, 2021
Claim/ View/Print
Margaret Hamilton
October 25, 2021
Claim/ View/Print
Joan Clarke
October 20, 2021
Claim/ View/Print
Scripts Received

7.4 References

AI - Grok